

Material-aware rapid prediction of low-velocity impact damage in laminated composites via surrogate modeling

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Abstract

Low-velocity impact (LVI) can leave laminated composites with internal damage that is difficult to observe visually but may still degrade structural integrity. Finite-element simulation can resolve such damage fields, yet repeated case-specific analyses become costly when impact direction, ply orientation, stacking sequence, and material system vary together. Existing surrogate models often treat material information as a simple label and may overlook low-intensity regions in finite-element-generated fiber damage fields. This study develops a material-aware surrogate framework for rapid prediction of finite-element-generated fiber damage fields. The framework learns compact representations of damage images and maps physical parameters, including impact angles, ply position, ply orientation, and material type, to fiber tensile and fiber compressive damage fields. Validation shows that the model preserves global damage morphology, captures fiber orientation-induced rotation, represents symmetry loss under oblique impact, and adjusts damage intensity across material systems under the controlled lay-up setting considered in this work. With an inference time of approximately 5 ms, the framework provides a fast alternative to repeated finite-element analysis for preliminary impact damage evaluation and design screening.

Keywords

Laminated composites, low-velocity impact damage, material-aware, surrogate modeling

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Introduction

Laminated composites offer exceptional mechanical performance and design versatility through tailored parameters like ply orientation, stacking sequence, and material selection (Lei et al., 2024; Minigher et al., 2024; Peng et al., 2025). For instance, the strengthening or weakening effect of specific orientations has been systematically quantified (Peng et al., 2025), while the role of stacking sequence in governing fatigue damage evolution has been clarified (Lei et al., 2024), alongside sensitivity analyses that identify the dominant material parameters for intralaminar damage prediction (Minigher et al., 2024). This design freedom enables precise control of stiffness (Sheikhi et al., 2024), strength (Wang et al., 2024c), and damage tolerance (Sebaey et al., 2013) for diverse applications spanning automotive, aerospace, and marine structures (Choudhary et al., 2026; Edwin Raja Dhas et al., 2023; Gao et al., 2024; Garnier et al., 2011; Katnam et al., 2013). Building on these advantages, laminated composites nevertheless face limitations in impact resistance due to weak interlaminar interfaces, where property degradation under fatigue loading becomes a key concern (Guo et al., 2025; Khan et al., 2024). When subjected to impact loading, they develop complex damage mechanisms including macroscopic delamination and matrix cracking (Cantwell and Morton, 1989; Flores et al., 2017; Othman et al., 2016), and barely visible impact damage (BVID)-like low-intensity damage regions (Bergan et al., 2012). In this study, this term refers to weak regions in finite-element (FE)-generated fiber damage fields rather than experimentally certified BVID. Similar visually hidden damage can reinitiate and grow under subsequent loading (Jribi et al., 2024). Such damage can compromise stiffness (Lai et al., 2021) and in-plane compressive strength (Lai et al., 2019) of laminated composites; postimpact compressive strength degradation has also been explored through data-driven prediction methods (Hasebe et al., 2025). Accurate prediction of such impact-induced damage poses a fundamental challenge for ensuring composite structural integrity, particularly in safety-critical applications where hidden damage progression may lead to catastrophic failure without visible warning signs.

Numerical simulation is widely used to analyze low-velocity impact (LVI) damage in laminated composites (Gong et al., 2022; Katunin et al., 2021; Sommer et al., 2022), as it supports flexible parametric modeling and direct visualization of damage fields. Simulation outcomes depend on the selected damage criterion. The Hashin criterion models in-plane fiber and matrix damage (Hashin, 1980) but does not account for the through-thickness failure modes needed in impact analysis. The Puck criterion improves matrix compression and interfacial failure prediction through its fracture plane approach (Puck and Schürmann, 2002), whereas the Langley Research Center (LaRC) criterion enables full three-dimensional (3D) stress coupling at higher computational cost (Davila et al., 2005; Pinho et al., 2005). Among these, the 3D-Hashin criterion has gained prominence in contemporary research. For instance, Trombini et al. (2024) employed a full 3D-Hashin orthotropic damage model, which includes through-thickness stress components in the damage criterion, within a one-dimensional (1D) higher-order theoretical framework for quasi-static progressive failure analysis. Despite these advantages, numerical simulation is still case specific. Unsymmetric stacks, hybrid laminates, other ply configurations, and impact conditions defined by angle and velocity each require customized FE models. This becomes costly during stacking sequence optimization or sensitivity studies, where repeated model construction limits design exploration. These costs motivate material-aware surrogate modeling methods that keep material and impact parameters explicit while enabling rapid prediction. Recent studies have also combined LVI experiments, numerical models, and machine learning to characterize damage and postimpact response across laminate configurations (Koçak et al., 2026; Liu et al., 2025; Masud et al., 2026; Sommer et al., 2025; Yue et al., 2026; Zhao et al., 2026).

Surrogate models can predict damage rapidly when the mapping between inputs and damage fields is highly nonlinear (Bhaduri et al., 2022; Chen et al., 2023; Lee et al., 2025; Li et al., 2023; Ren et al., 2025; Rezasefat et al., 2024; Soar et al., 2024; Song et al., 2025; Tang et al., 2024a, 2024b; Tsai et al., 2023; Wang et al., 2024a, 2024b; Xiang et al., 2024; Xu et al., 2024; Yang et al., 2021; Zhao et al., 2026).

Generative models are useful for predicting stress and strain fields in composites, as shown by Gupta et al.'s U-Net architecture for fiber-reinforced composite stress field prediction (Gupta et al., 2023) and Wang et al.'s generative adversarial network (GAN)-based model for strain distributions in concrete composites under compression (Wang et al., 2023). In composites, stress, strain, and damage fields are spatially linked: stress concentrations coincide with damage initiation sites, and damage paths follow regions with high stress gradients (Lu et al., 2024). Damage evolution adds nonlinear effects that are not present in stress-strain analysis, including progressive stiffness degradation and discontinuous delamination propagation (Shi et al., 2012). These input-output relationships are suitable for high-fidelity surrogate models that can learn high-dimensional, physics-coupled mappings (Gupta et al., 2023; Wang et al., 2023). Recent frameworks have also modeled impact-damage relationships under different LVI conditions (Wang et al., 2024a). Yet most surrogate models still treat material information weakly and give limited attention to BVID-like low-intensity damage regions in FE damage images. Such models can reduce computational cost, but material-aware prediction is needed when the same damage shape corresponds to different material thresholds. Recent surrogate and segmentation studies further show that physics constraints, phase-field data, and microstructural image features can be embedded in learning frameworks for mechanics-relevant field prediction (Zhang et al., 2025a, 2025b, 2026a, 2026b; Zhou et al., 2025). Related studies on cross-material fatigue, autoregressive inverse design, and physics-guided diffusion further show how material descriptors can separate shared response trends from material-specific shifts (Xie et al., 2026; Zhang et al., 2025a; Zhao et al., 2026).

These physical correlations and nonlinear mapping capabilities support the extension of stress and strain prediction models to damage field analysis. However, three problems still limit impact-induced damage prediction for laminated composites, especially when the target includes material-dependent thresholds and BVID-like low-intensity damage regions:

1. *Limited damage representation*: Existing datasets contain too few BVID-like low-intensity damage regions, which reduces prediction accuracy in weakly damaged areas.
2. *Architectural limitations*: Current surrogate networks do not fully capture the high-dimensional material relationships that govern damage fields, leading to performance degradation on complex stacking sequences and weak generalization across material systems.
3. *Gap in integration of material-aware information*: Current models do not combine impact dynamics with laminate characteristics, including ply angles and material types. This limits their ability to learn material-specific damage thresholds and preserve BVID-like low-intensity damage regions across material configurations.

To address these challenges, this study develops a material-aware surrogate model with a feature-fusion architecture. The framework builds a dataset containing minor fiber damage features and uses a dual-branch structure to combine external impact parameters with internal laminate characteristics. In the present work, the learning target is restricted to FE-generated fiber tensile damage (TD) and fiber compressive damage (CD) fields; matrix damage and delamination-specific variables are not used as prediction targets. By generating damage fields directly from physical parameters, the model supports efficient impact-damage assessment and analysis of fiber damage morphology under LVI.

Parametric modeling

FE models were developed in Abaqus/Explicit to simulate LVI on laminated composites, generating the numerical dataset required for surrogate model training. The progressive damage behavior was modeled through a VUMAT subroutine implementing the 3D-Hashin failure criterion, which served as the constitutive basis for material degradation.

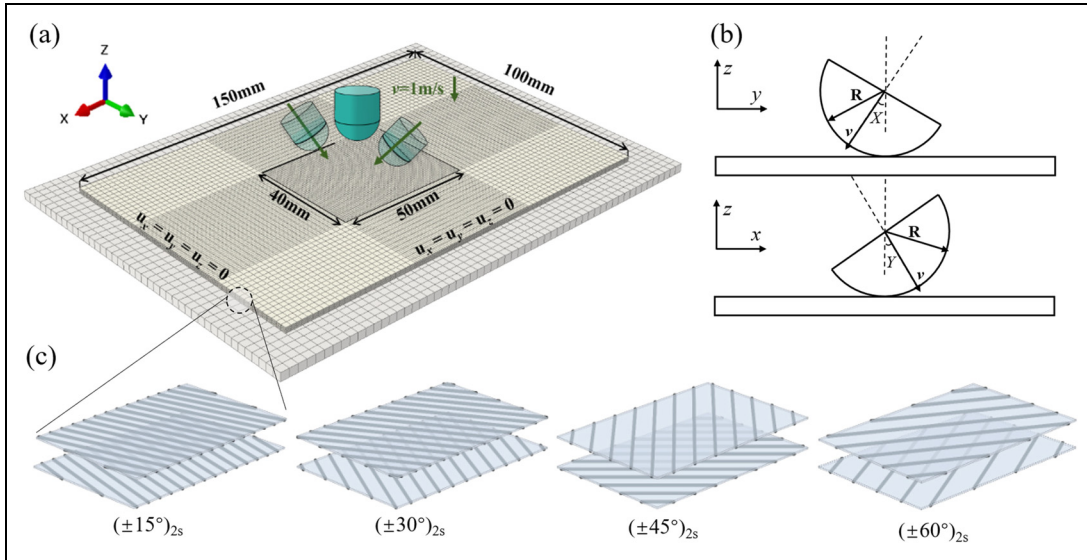


Figure 1. Finite-element (FE) model setup. (a) Mesh-configured composite laminate model for LVI simulation; (b) bidirectional impactor velocity loading; (c) and stacking sequence design.

As shown in Figure 1(a), the FE impact model was developed with reference to work carried out by Huang et al. (2023). The model consisted of a laminated composite, a rigid impact body, and a rigid support fixture. The laminated composite had dimensions of 150 mm $\times$ 100 mm $\times$ 2 mm and incorporated multiple ply orientation configurations. The laminate model utilized a refined central mesh with an element size of 0.8 mm, while nonrefined regions maintained a mesh size of 2.0 mm, as displayed in Figure 1(a). The laminate was discretized by eight plies of C3D8R hexahedral elements with 64,640 elements in total. To represent interlaminar interaction, seven surface-to-surface contact pairs were defined between adjacent plies through the thickness. Each interface used hard normal contact with separation allowed and penalty-based tangential friction under finite sliding. The present model did not adopt a cohesive surface technique, cohesive elements, or an independent interface damage evolution law. This contact-based setting was used to keep the FE data generation procedure consistent and time-efficient across all material systems in the present surrogate modeling study. A more detailed delamination formulation based on cohesive interfaces is outside the scope of this work. A stepwise summary of the interlaminar contact definition is provided in Appendix Algorithm A1.

A Python script-based parametric modeling program was developed to enable automated batch generation of FE models. As shown in Figure 1(c), the laminate lay-up, as one of the design parameters, was modeled in FE analysis with stacking sequence covering $(0^\circ)_8$, $(\pm 15^\circ)_{2s}$ to $(\pm 75^\circ)_{2s}$, and $(90^\circ)_8$, employing 15° increments for angle specification. In Figure 1(a), a hemispherical impactor with a radius of 8 mm engages with the laminate's top surface center through multiangle orientations. As illustrated in Figure 1(b), the direction angles of the velocity vector v were defined by (1) clockwise rotation angle X about the x -axis and (2) rotation angle Y about the y -axis, defining an oblique-impact velocity vector space. A concentrated mass of 5 kg was assigned to the reference point located at the impactor's crown, with a constant initial velocity of 1 m/s imposed along the prescribed impact direction.

This lay-up set was selected as a controlled orientation sweep rather than a direct reproduction of the most common laminate design library. The 0° , 45° , and 90° directions are standard building orientations

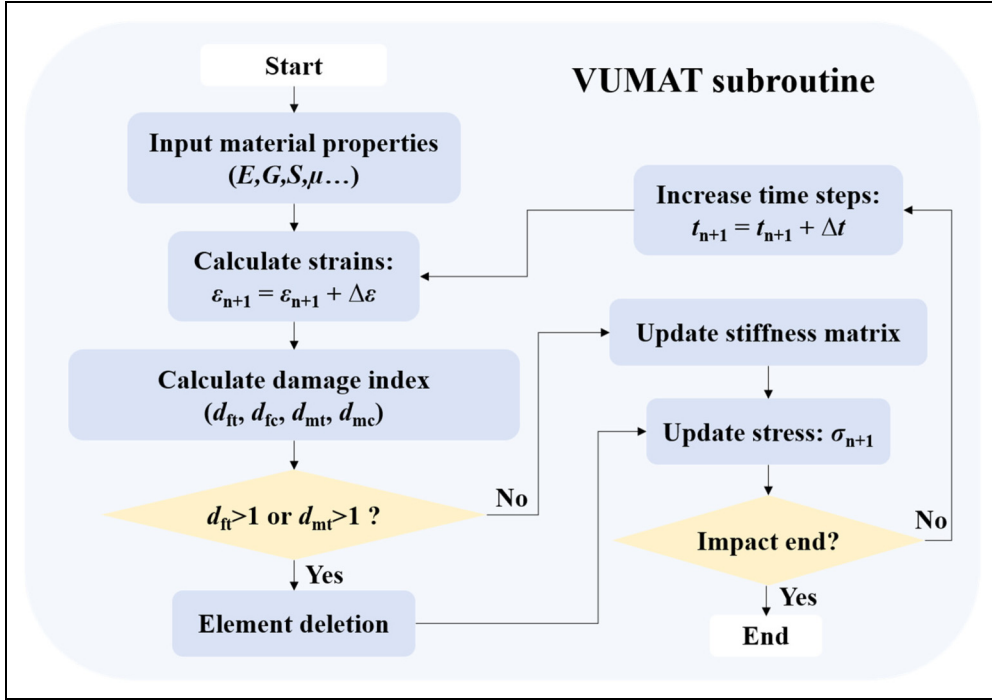


Figure 2. Implementation flowchart of the VUMAT subroutine. 3D-Hashin-based damage index calculation with dynamic element deletion for laminated composite impact simulation. 3D: three-dimensional.

in laminate design, and quasi-isotropic laminates are commonly formed by combining 0° , $\pm 45^\circ$, and 90° plies (Cameron et al., 2023). Meanwhile, orthogonal $0^\circ/90^\circ$ lay-up concepts are still used in specific engineering structures, such as composite risers (Amaechi, 2022). Therefore, the present design kept these orientations as boundary and reference cases and used 15° increments to provide a structured input space for surrogate model training.

Progressive damage analysis was implemented through a user-defined material (VUMAT) subroutine, as depicted in Figure 2, where the 3D-Hashin failure criterion (Trombini et al., 2024) served as the constitutive basis. The VUMAT governs ply-level stiffness degradation and element deletion during impact. Therefore, the FE model captured ply-level progressive damage together with interlaminar contact and possible separation, but it did not introduce a separate delamination damage variable, delamination initiation criterion, or delamination propagation law. This modeling choice provided a clear and unified laminate damage description for the present surrogate modeling study, while a separate interface damage formulation was not considered in this work. The constitutive equations are defined as:

$$d_{ft} = \left(\frac{\varepsilon_{11}}{\varepsilon_{1t}} \right)^2 + \left(\frac{\gamma_{12}}{\gamma_{12}^{ult}} \right)^2 + \left(\frac{\gamma_{13}}{\gamma_{13}^{ult}} \right)^2 \geq 1 \quad (1)$$

$$d_{fc} = \left(\frac{\varepsilon_{11}}{\varepsilon_{1c}} \right)^2 \geq 1 \quad (2)$$

Table 1. Parameters for FE modeling of LVI on laminated composites.

Material type	Stacking sequence	Velocity directions (X or Y)	Number of FEM
T700GC/M2I (Li et al., 2024)	$(0^\circ)^8, (\pm 15^\circ)_{2s} \dots (\pm 75^\circ)_{2s}, (90^\circ)^8$	$10^\circ, 20^\circ \dots 80^\circ$	448
AS4/8552 (Falcó et al., 2022)	$(\pm 30^\circ)_{2s}$	$10^\circ, 20^\circ \dots 80^\circ$	64
CFRP (Shi et al., 2012)	$(\pm 30^\circ)_{2s}$	$10^\circ, 20^\circ \dots 80^\circ$	64
T700/790I (Hou et al., 2024)	$(\pm 30^\circ)_{2s}$	$10^\circ, 20^\circ \dots 80^\circ$	64

FE: finite element; FEM: finite-element model.

$$d_{mt} = \left(\frac{\epsilon_{22}}{\epsilon_{2t}} \right)^2 \geq 1 \quad (3)$$

$$d_{mc} = \frac{(\epsilon_{22} + \epsilon_{33})^2}{\epsilon_{2c}\epsilon_{3c}} + \frac{\epsilon_{22} + \epsilon_{33}}{\epsilon_{2c}} \left(\frac{f_{2c}}{2G_{12}} - 1 \right) + \left(\frac{\gamma_{12}}{\gamma_{12}^{ult}} \right)^2 + \left(\frac{\gamma_{13}}{\gamma_{13}^{ult}} \right)^2 + \left(\frac{\gamma_{23}}{\gamma_{23}^{ult}} \right)^2 \geq 1 \quad (4)$$

where ϵ_{11} denotes the fiber-direction normal strain; γ_{12} and γ_{13} correspond to in-plane and out-of-plane shear strain components associated with fiber–matrix interaction; ϵ_{22} indicates the transverse normal strain in the matrix phase; ϵ_{33} quantifies through-thickness normal strain; and γ_{23} characterizes the transverse shear stress at the ply interface. The stress/strain components with the superscript *ult* are their limiting values.

To check the numerical reliability of the FE reference model, a mesh sensitivity analysis is added in Appendix A7. The contact-force histories obtained with different mesh sizes show that the adopted mesh setting gave a response close to the finer mesh during the main loading stage, while coarser meshes led to larger deviations in the peak force and later-stage response. Direct experiment-based validation for the present material systems is outside the scope of this work, which focuses on surrogate-model construction using FE reference fields.

The established parametric modeling framework integrated two aspects: internal laminate properties, including stacking sequences and material systems, and external impact dynamics, described by the velocity direction angles (*X*, *Y*). The angles *X* and *Y* were discretized in 10° increments from 10° to 80° , while the lay-up family was defined as a controlled set of symmetric laminates with orientation angles from 0° to 90° at 15° intervals. This lay-up design gave broad but structured coverage for surrogate-model training rather than a full engineering laminate library. Accordingly, the 0° and 90° cases were retained as boundary orientations, and the independent 45° ply validation set was used as the main cross-ply generalization test. The main training dataset listed in Table 1 did not include a quasi-isotropic lay-up. A small additional unseen-sequence validation case is introduced in Appendix A6, but quasi-isotropic coverage is not included in the present dataset. This configuration generated 448 FE model variants under one material type. For each additional material type, 64 parameter combinations were created using a fixed $\pm 30^\circ$ ply configuration, forming an FE simulation dataset containing 640 cases, as shown in Table 1. The elastic constants, strength parameters, and density values used for the four material systems are listed in Appendix Table A1.

Deep learning framework for impact damage prediction

As illustrated in Figure 3, the proposed material-aware surrogate model integrates two computational branches to resolve the high-dimensional mapping challenge between scalar impact parameters and complex spatial damage fields. In the present study, the predicted spatial damage fields specifically refer to fiber TD fields and fiber CD fields, which are learned in separate training tasks. The framework follows a two-stage logic to bridge the significant dimensional discrepancy between scalar impact parameters and spatial

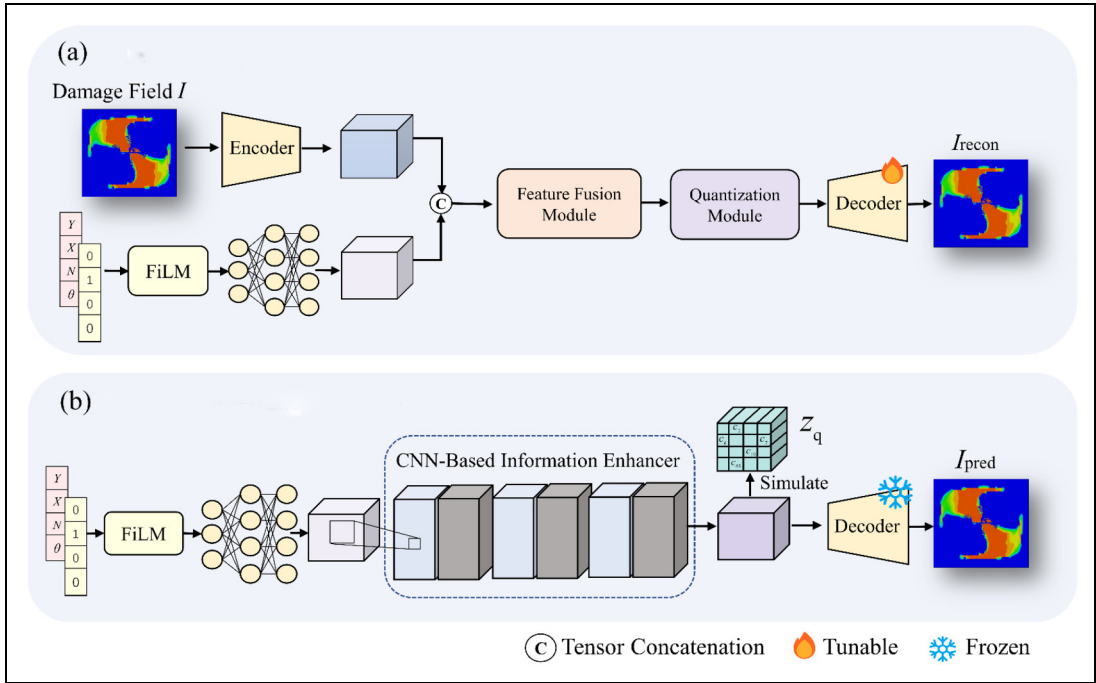


Figure 3. Architecture of the material-aware surrogate model. (a) Damage image reconstructor and (b) parameter-to-image predictor.

damage distributions. First, a damage image reconstructor (DIR) was developed to compress various damage morphologies into a representative latent space that captured structural anisotropy and material-specific features. Second, a parameter-to-image predictor (PIP) was trained to map external impact conditions and internal laminate characteristics to this learned latent space. During the inference phase, the model only requires the PIP to take physical parameters as input and directly generate damage field images through the pretrained decoder, enabling rapid evaluation without the original damage images.

Damage image reconstructor

DIR pipeline. Input data to DIR consisted of three components: processed damage images I_i , parameter vectors v_i , and material descriptors e_i . For the i -th sample, each damage image I_i is a two-dimensional image representation of the FE damage field for one simulation case. This FE damage field was extracted from the postimpact results of Abaqus/Explicit and mapped onto the laminate domain through pixel intensities using the normalized damage index, where a pixel value close to 0 denotes an undamaged state and a value close to 1 denotes severe damage with possible element deletion. Therefore, I_i is a direct image-form representation of the FE damage field rather than a schematic illustration or a time-resolved damage sequence. In the present work, this static image representation was adopted as a focused validation setting for surrogate-model training, while time-resolved damage imaging was not considered.

The parameter vector consisted of the impact direction angles defined in the “Parametric modeling” section, the ply sequence number through the laminate thickness, and the fiber orientation angle of the ply corresponding to the target damage image. In the intermaterial setting, a one-hot vector identifies the four

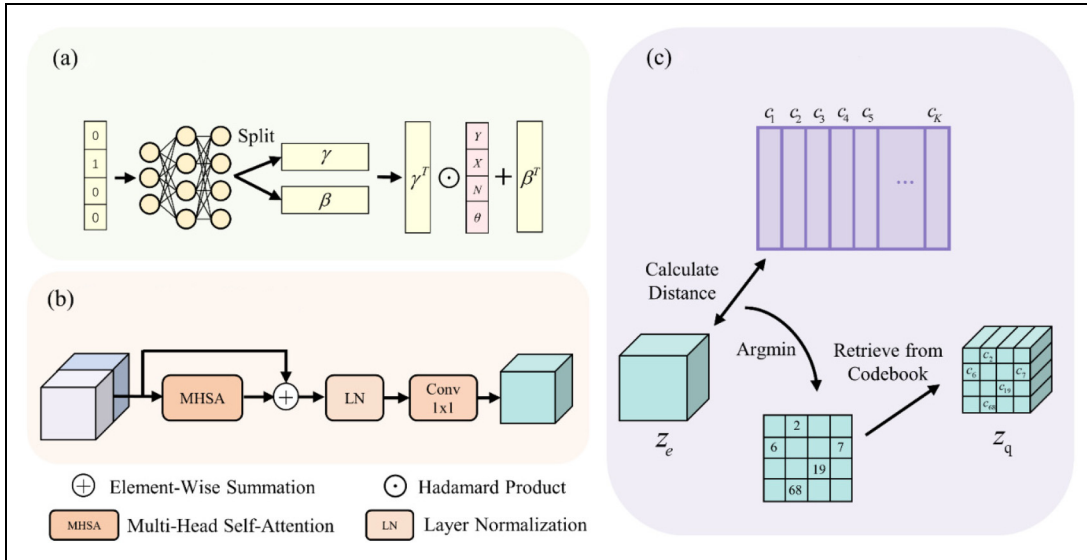


Figure 4. Illustration of data flow in the modules used in the material-aware surrogate model. (a) Feature-wise linear modulation module (FiLM); (b) feature fusion module; and (c) quantization module.

material systems listed in Table 1, namely T700GC/M21, AS4/8552, carbon fiber-reinforced polymer (CFRP), and T700/7901, whereas in the intramaterial setting, e_i is omitted because only one material system is considered. Given I_i , the encoder produced a latent feature $z_e \in \mathbb{R}^{C \times H \times W}$, where C denotes the number of channels in the last convolutional layer, and H and W denote the spatial size of the latent feature map. The Feature-wise Linear Modulation (FiLM) module uses e_i to modulate v_i , and the auxiliary multilayer perceptron (MLP) reshapes the modulated parameter features into the same form so that they can be fused with z_e in the Feature Fusion Module. The fused latent representation was then quantized and decoded to reconstruct the damage image.

The overall DIR backbone remains the same in both the intramaterial and intermaterial settings. The only difference lies in the material-conditioning path. In the intermaterial setting, the one-hot material descriptor e_i is used in the FiLM module to modulate v_i before feature reshaping and fusion. In the intramaterial setting, since only one material system is considered, e_i and the FiLM-based conditioning branch are omitted, and v_i is directly passed to the auxiliary MLP before entering the same downstream Feature Fusion Module, quantization module, and decoder. This design keeps the encoder-decoder architecture consistent across the two settings while introducing material awareness only when cross-material prediction is required.

FiLM module. The FiLM module was employed to model the nonlinear modulation of material properties on damage thresholds, providing representational capacity compared to simple feature concatenation (Perez et al., 2018). This integration was essential because the damage morphology is codetermined by the impact momentum components induced by the oblique impact angles and the constitutive limits of the material. Here, momentum partitioning refers to resolving the prescribed impactor momentum with respect to the laminate surface into a through-thickness normal component and in-plane tangential components. As Figure 4(a) illustrates, the modulation coefficients were obtained by transforming e_i . To address the sparsity of e_i characterized by a high number of zero values, GELU was utilized as the activation function.

The process is given as follows:

$$H = W_3 \cdot \text{GELU}(W_2 \cdot \text{GELU}(W_1 e_i^T)) \quad (5)$$

$$\gamma, \beta = \text{Split}(H, \dim = 1) \quad (6)$$

$$M = \gamma \odot v + \beta \quad (7)$$

where $\odot$ represents the Hadamard product, and W_i represents the weights of the i -th layer ($i = 1, 2, 3$). Then, the prefused data M were processed using an auxiliary MLP shown in Figure 3(a) and (b) after the FiLM module. In order to ensure adequate fusion afterward, data M were uniformly mapped to the range of $[0, 1]$ after Sigmoid and reshaped to obtain $\mathcal{F}_{\text{vector}}$.

Feature fusion module. To capture anisotropic energy dissipation paths and fiber-guided damage propagation in laminated structures, the Feature Fusion module uses an attention-based fusion mechanism. In this work, the same fusion step also carried the material-aware role: it placed material descriptors, impact parameters, and damage-image features in a shared representation before quantization. Attention-based fusion mechanisms have shown useful performance in integrating heterogeneous data types (Jiang et al., 2026; Truong et al., 2025; Zhang et al., 2025a, 2026a; Zhang and Tang, 2025), providing a technical basis for damage prediction in composite materials under complex impact scenarios. As shown in Figure 4(b), a multihead self-attention architecture (Vaswani et al., 2017) was used to model interactions between parameter features and damage-image features. This let the network connect fiber orientation with high-gradient damage zones and connect material descriptors with damage-threshold changes. The fusion process began by concatenating the encoder feature and the reshaped parameter feature along the channel dimension to form a fused feature map. This combined representation was projected through linear transformations to generate query (Q), key (K), and value (V). To improve computational efficiency, we replaced feed-forward networks with Layer Normalization while preserving information flow through residual connections (Ba et al., 2016; He et al., 2016). The workflow maintained gradient stability and gave the latent space enough separation to represent material-specific damage responses, including weak low-intensity morphology as well as larger damage zones. Recent top-conference work further suggests a shift from generic attention blocks toward fine-grained or modular fusion when heterogeneous features carry complementary information (Li et al., 2025; Yu et al., 2025):

$$G_i = \text{Softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i \quad (8)$$

$$z_e = \text{Conv}_{1 \times 1}(\text{LayerNorm}(G + \mathcal{F})) \quad (9)$$

where i denotes the i -th head and d_k denotes the dimension of one head.

Quantization module. This module performs the vector quantization operation of the Vector-Quantized Variational Autoencoder (VQ-VAE) architecture to discretize the continuous latent space (van den Oord et al., 2017). Here, the codebook is a learnable embedding table in the latent space, and each codebook vector is a prototype latent vector. In the present implementation, the codebook contained 512 learnable vectors, and each vector had dimension 16. As shown in Figure 4(c), for the fused latent feature vector at each spatial location in the encoder output, the model computed its distance to all codebook vectors and replaced

it with the nearest one to obtain the quantized latent representation. The quantization process is formulated as follows:

$$k = \underset{k \in \{1, \dots, K\}}{\operatorname{argmin}} \|z_e^{(ij)} - c_k\|_2^2 \quad (10)$$

$$z_q^{(ij)} = c_k \quad (11)$$

where $\|\cdot\|_2$ denotes the L_2 norm. Then the quantized latent features z_q are transformed through the Decoder into the reconstructed damage images I_{recon} .

Loss function of DIR. The loss function of DIR consists of three components: reconstruction loss, commitment loss, and codebook loss. The commitment loss constrains the encoder output to remain close to the selected codebook vectors, while the codebook loss updates the learnable codebook embeddings. Following the standard VQ-VAE formulation (van den Oord et al., 2017), these embeddings were optimized by backpropagation through the codebook loss. The stop-gradient operator prevents this loss term from updating the encoder output, so its gradient was applied only to the selected codebook embeddings. The loss is formulated as follows:

$$\mathcal{L}_{\text{recon}} = \|I - I_{\text{recon}}\|_2^2 \quad (12)$$

$$\mathcal{L}_{\text{codebook}} = \|sg(z_e) - z_q\|_2^2 \quad (13)$$

$$\mathcal{L}_{\text{commit}} = \|z_e - sg(z_q)\|_2^2 \quad (14)$$

$$\mathcal{L}_{\text{DIR}} = \mathcal{L}_{\text{recon}} + \alpha \cdot \mathcal{L}_{\text{codebook}} + \beta \cdot \mathcal{L}_{\text{commit}} \quad (15)$$

where hyperparameters $\alpha = 1.0$ and $\beta = 0.25$, $sg[\cdot]$ denotes the stop-gradient operator.

Parameter-to-image predictor

PIP pipeline. The PIP pipeline serves as the inference engine that transforms low-dimensional parameters into spatial damage distributions. The PIP backbone is identical in the intramaterial and intermaterial settings, while the material-conditioning path differs. In the intermaterial setting, FiLM is employed to integrate the feature vector v with the material-type encoding e_i before latent prediction. In the intramaterial setting, this material-conditioning path is removed, and v is directly fed to the downstream predictor. To address the limitation of pure MLP in low-to-high dimensional mapping, existing research demonstrates that integrating convolutional neural networks (CNNs) with traditional MLP frameworks can enhance representational capacity (Ding et al., 2021). We accordingly designed a CNN-based Information Enhancer (CIE), utilizing LeakyReLU activation to prevent neuron deactivation. In both settings, the surrogate model output latent features through dimensional mapping and reshaping operations, optimized to closely approximate the quantized features generated by the pretrained DIR encoder. This alignment allowed the PIP to predict spatial morphology using only physical parameters during the inference phase. The final damage field images were predicted via a frozen-weight DIR decoder.

Loss function of PIP. To enable PIP to achieve high-quality prediction, we designed two loss functions: the latent loss and the prediction loss, which constrain the imitation of quantized latent features and the prediction maps, respectively, formulated as follows:

$$\mathcal{L}_{\text{latent}} = \|z_L - z_q\|_2^2 \quad (16)$$

$$\mathcal{L}_{\text{pred}} = \|I_{\text{pred}} - I_{\text{recon}}\|_2^2 \quad (17)$$

$$\mathcal{L}_{\text{PIP}} = \alpha \cdot \mathcal{L}_{\text{latent}} + \beta \cdot \mathcal{L}_{\text{pred}} \quad (18)$$

where hyperparameters $\alpha = \beta = 1.0$ balance the two loss terms.

Data processing and training configuration

Dual-stage validation. This study established two progressive experimental phases to validate the performance and generalization capabilities of the material-aware surrogate model in both intramaterial and intermaterial impact damage prediction scenarios. The intramaterial phase focuses on evaluating foundational performance restricted to one material type through hierarchical validation strategies, ensuring robustness in intramaterial generalization while isolating material-specific variables. The intermaterial phase extends this framework to assess feature fusion mechanisms for capturing minor damage morphologies across different material types, leveraging asymmetric data organization to mitigate data bias and validate intermaterial generalizability. Both phases employed the same strategy, normalized to $[0, 1]$ via equations (19) and (20), and a training configuration using an NVIDIA RTX 4090 GPU with learning rate scheduling governed by equations (21) and (22), featuring modular adjustments aligned with phase-specific objectives.

For reproducibility, the intramaterial and intermaterial settings were trained separately, with no parameter sharing or cross-setting fine-tuning. Fiber TD and fiber CD were also modeled independently. Each damage type had its own DIR and matched PIP, and the two model pairs did not share parameters. For each task, the DIR was trained first to learn the latent representation and decoder, followed by the PIP trained against the latent targets produced by the corresponding DIR:

$$v_{\text{norm}} = \frac{v - v_{\min}}{v_{\max} - v_{\min}} \quad (19)$$

$$I_{\text{norm}} = \frac{I}{255} \quad (20)$$

$$lr_{\text{DIR}} = \begin{cases} 1 \times 10^{-3}, & 0 \leq \text{epoch} \leq 200 \\ 1 \times 10^{-4}, & \text{otherwise} \end{cases} \quad (21)$$

$$lr_{\text{PIP}} = \begin{cases} 1 \times 10^{-3}, & 0 \leq \text{epoch} \leq 500 \\ 1 \times 10^{-4}, & \text{otherwise} \end{cases} \quad (22)$$

Intramaterial damage field prediction phase. For intramaterial evaluation, the surrogate model was separately trained on two target datasets, namely fiber TD and fiber CD, using parameter vectors v and damage images I . The damage images were fiber TD or fiber CD fields extracted from the final time step of each 0.01 s FE simulation. At this time, the impactor had entered the rebound stage in nearly all the cases, and the damage state was close to its maximum, providing a consistent extraction point across the dataset. The present framework is not restricted to fiber damage as a modeling architecture. If matrix TD fields or matrix CD fields are prepared in the same form, the same framework can be trained for those targets as well. However, the present paper validated only fiber-damage prediction; matrix-damage prediction and fully time-resolved evolution were not considered.

Here, Y and X represent impact angles, N represents the ply sequence number, and θ represents the varying stacking sequence. The intramaterial data were organized into one training split and two held-out

validation subsets. The independent 45° ply validation set was entirely excluded from gradient-based training and therefore did not participate in backpropagation or weight updates. The remaining non-45° ply data were partitioned into the training split and the in-plane validation subset with an 8:2 ratio. Here, “in-plane” refers to the set of fiber-orientation angles defined within the laminate plane, and the validation subset consisted of held-out samples from the non-45° configurations. During model development, hyperparameter selection and checkpoint comparison were conducted according to the validation performance on the held-out validation subsets, including both the 45° ply validation set and the in-plane validation set, so that the selected configuration remained effective for same-ply prediction and cross-ply generalization.

Architecturally, the intramaterial model retained the same DIR and PIP backbones as the intermaterial model while removing only the material-conditioning branch. Since all samples in this setting belong to one material system, the one-hot material descriptor e_i was not introduced, and FiLM was not activated. Accordingly, in DIR the parameter vector v was directly transformed by the auxiliary MLP before fusion with the encoder features, and in PIP the same predictor and CIE modules operated using only v .

Intermaterial damage field prediction phase. This experiment couples impact dynamics parameters, stacking sequence, and material types, building upon the previously validated intramaterial experimental performance. The same fiber-damage-field definition of I was used in the intermaterial phase. With previous research confirming the effectiveness of one-hot encoding for material representation (Srinivasan et al., 2024), material types were parameterized using an expandable one-hot encoded vector. The asymmetric data organization strategy was adopted as follows: the dataset allocated impact damage field images from four distinct laminated composites in a 7:1:1:1 ratio for intermaterial analysis while maintaining an 8:2 training-to-validation split within each material-specific subset. Architecturally, the intermaterial model retained the same DIR and PIP backbones as the intramaterial model, but activated the FiLM-based material-conditioning branch to incorporate e_i into both reconstruction and prediction. Reconstruction inputs therefore included I , v , and e_i , while prediction uses v and e_i exclusively, as shown in Table 2.

Results and discussion

Analysis of low-velocity impact damage mechanisms in laminated composite

Comparative analysis of the fiber-damage states obtained from Abaqus simulations reveals parametric sensitivity under LVI, as displayed in Figure 5. In Figure 5, the contours labeled Tensile Damage and Compressive Damage correspond to fiber TD and fiber CD, respectively, rather than matrix-damage fields. Figure 5 does not show a separate delamination pattern. Matrix and delamination damage are not considered in this work as mentioned earlier. Instead, it compares representative fiber-damage fields extracted at specific analysis increments under different impact conditions, mainly from the later stage where the damage state becomes pronounced. The influence of impact velocity direction on fiber-damage distribution is non-linear. When the velocity direction deviates from the normal, the reduced normal component leads to a smaller overall damage area. The oblique impact also breaks the center symmetry of the damage field and shifts the damaged region toward the upper part of the layer-level map. This in-plane localization is particularly evident in layer 7. This pattern suggests that impact momentum direction helps shape fiber-damage morphology in these representative cases.

As observed in cases (a) and (b), keeping the impact velocity direction fixed while changing the stacking sequence from 30° to 60° rotates the damage morphology. This change is consistent with a different stress-transfer path induced by ply orientation. In this representative comparison, the 60° case shows lower fiber compressive damage in the mid-layers (layers 4–6) than the 30° case. This trend may be associated with the changed anisotropic stiffness distribution and stress-transfer path induced by the ply orientation.

Table 2. Comparison of training protocols and model configurations for intramaterial and intermaterial settings.

Aspect	Intramaterial setting	Intermaterial setting
Backbone architecture	DIR and PIP backbones are the same as those in the intermaterial setting.	DIR and PIP backbones are the same as those in the intramaterial setting.
Material conditioning	FiLM and e_i are omitted.	FiLM is activated and uses e_i .
Reconstruction and prediction inputs	DIR uses l and v_i . PIP uses v_i only.	DIR uses l , v_i , and e_i . PIP uses v_i and e_i .
Data split	Non-45° data are split into training and in-plane held-out validation with a ratio of 8:2.	Each material subset is split into training and held-out validation with a ratio of 8:2 after the 7:1:1:1 material organization.
Role of 45° data	All 45° ply cases are fully excluded from training and used only as an independent held-out validation subset.	No dedicated 45° held-out subset is defined.
Test set	No separate test set is introduced.	No separate test set is introduced.
Model selection	Hyperparameters and checkpoints are selected from held-out validation performance. The 45° subset is not used in backpropagation.	Hyperparameters and checkpoints are selected from held-out validation performance.
Training relation	Intramaterial and intermaterial models are trained separately and independently. Fiber tensile and fiber compressive damage are also trained as separate tasks.	Intramaterial and intermaterial models are trained separately and independently. Fiber tensile and fiber compressive damage are also trained as separate tasks.
Training order and epochs	DIR is trained first, then the matched PIP. DIR: 300 epochs. PIP: 1000 epochs.	DIR is trained first, then the matched PIP. DIR: 300 epochs. PIP: 1000 epochs.

DIR: damage image reconstructor; PIP: parameter-to-image predictor; FiLM: feature-wise linear modulation.

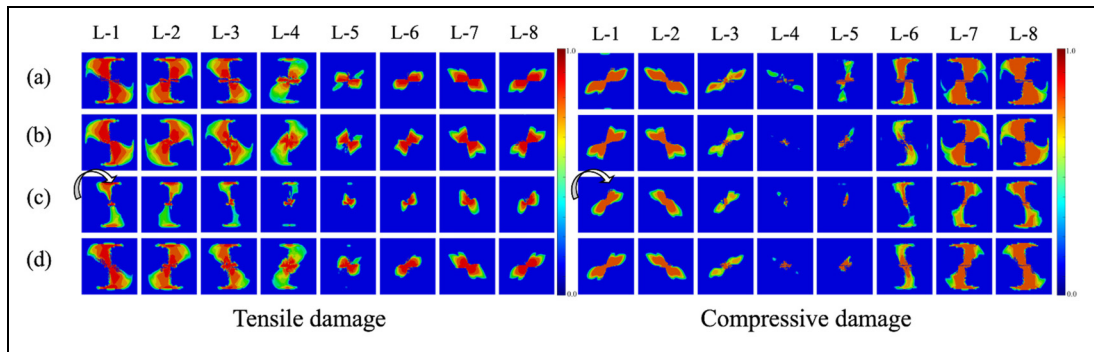


Figure 5. Representative fiber-damage fields extracted at the final time step ($t=0.01$ s) under different LVI conditions. (a) $X=10^\circ$, $Y=10^\circ$, $(\pm 30^\circ)_{2s}$, T700GC/M2I; (b) $X=10^\circ$, $Y=10^\circ$, $(\pm 60^\circ)_{2s}$, T700GC/M2I; (c) $X=60^\circ$, $Y=10^\circ$, $(\pm 30^\circ)_{2s}$, T700GC/M2I; and (d) $X=10^\circ$, $Y=10^\circ$, $(\pm 30^\circ)_{2s}$, T700/790I. The color bar represents the normalized damage index, where a value of 1.0 indicates complete material failure and subsequent element deletion based on the 3D-Hashin criterion, while 0.0 denotes the pristine, undamaged state. All damage fields were extracted at the final time step of the 0.01 s impact simulation, when the impactor was entering the rebound stage in nearly all the cases and the damage state was close to its maximum. The white curved arrows indicate the direction of damage morphology rotation from the top row to the bottom row within the marked columns. 3D: three-dimensional.

Material type is not the primary factor controlling the global damage skeleton in these representative cases, but it affects damage intensity and threshold. Under identical impact velocities and ply configurations, the selected materials show similar damage skeletons, as seen in Figure 5(a) and (d). The numerical results indicate that the global pattern is mainly governed by the structural lay-up, while material-specific properties shift the damage threshold and the external impact parameters influence the symmetry of the damage morphology, as seen in Figure 5(a) and (c).

Intramaterial damage field prediction via the material-aware surrogate model

Intramaterial results. This subsection evaluates the material-aware surrogate model for two separate intramaterial prediction tasks: fiber TD and fiber compressive damage CD. The evaluation first tests whether the model can preserve fiber-guided damage morphology within a fixed material system, including BVID-like low-intensity damage regions. These results provide the baseline for the later cross-material tests.

An error map visualizes the squared error between the true value and the predicted result at each pixel point. For an image of size $H \times W$, the squared error at each pixel (i, j) is calculated as:

$$\text{Error}_{(i,j)} = (y_{(i,j)} - \hat{y}_{(i,j)})^2 \quad (23)$$

where $y_{(i,j)}$ represents the ground-truth pixel value at position (i, j) , and $\hat{y}_{(i,j)}$ represents the predicted pixel value at position (i, j) , both of which are normalized grayscale values.

Root mean square error (RMSE) quantifies pixel-level discrepancies between predicted and ground-truth images, measuring numerical distribution deviations. For images X and Y , RMSE is calculated as:

$$\text{RMSE} = \sqrt{\frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W (X(i, j) - Y(i, j))^2} \quad (24)$$

where H and W denote image height and width. Lower RMSE values indicate closer alignment between predictions and ground truth.

Learned perceptual image patch similarity (LPIPS) (Zhang et al., 2018) is a deep perceptual metric that evaluates perceptual quality by computing differences in multilayer feature spaces of pretrained networks (e.g., Visual Geometry Group (VGG), AlexNet). For input X and prediction Y , LPIPS is calculated as:

$$\text{LPIPS}(X, Y) = \|\phi(X) - \phi(Y)\|_2^2 \quad (25)$$

where ϕ represents features extracted by the deep network. Lower LPIPS values reflect smaller perceptually discernible differences, aligning with human visual perception.

Structural similarity index (SSIM) (Wang et al., 2004) assesses structural similarity between images. For X and Y , SSIM is calculated as:

$$\text{SSIM}(X, Y) = \frac{(2\mu_X\mu_Y + c_1)(2\sigma_{XY} + c_2)}{(\mu_X^2 + \mu_Y^2 + c_1)(\sigma_X^2 + \sigma_Y^2 + c_2)} \quad (26)$$

where μ_X and μ_Y denote image means; σ_X and σ_Y represent variances; σ_{XY} is the covariance; and c_1 and c_2 are stabilization constants. SSIM ranges from 0 to 1, with higher values indicating greater structural similarity.

Due to the training set design, Figure 6(a) and (c) presents outcomes from the 45° ply-angle validation set, where damage fields are accurately predicted, with an error map showing that the 99th

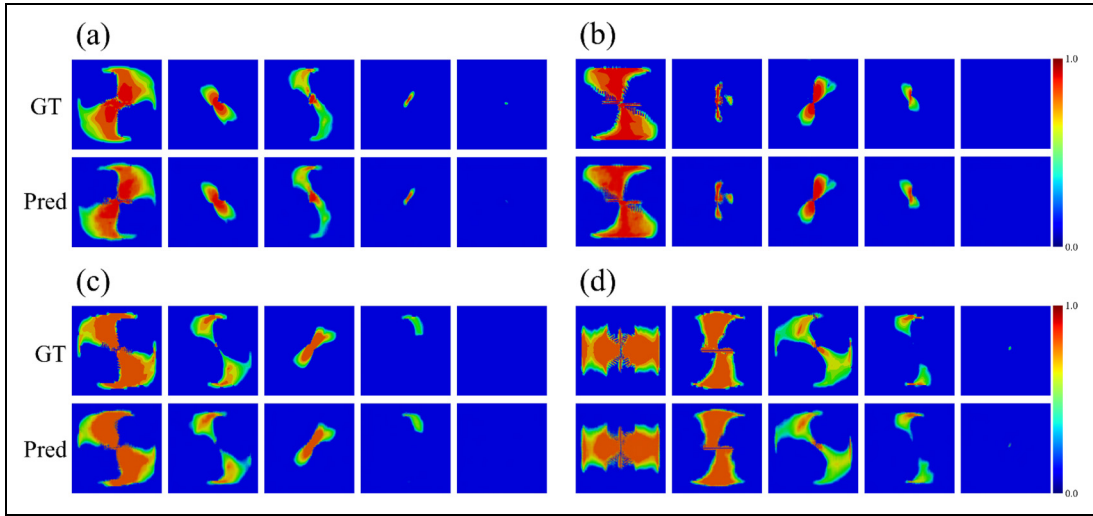


Figure 6. Validation set comparisons between ground-truth (GT) and predicted fiber-damage fields (Pred) in intramaterial experiments. Panels (a) and (b) show fiber tensile damage, while panels (c) and (d) show fiber compressive damage. The physical parameters of the five cases in each panel are listed in Table 3.

percentile squared error remains below 0.1 in cross-ply damage prediction tasks. In Figure 7, we employ the 99th percentile method to filter out the extreme errors in the top 1%. This method eliminates outlier interference and helps the model focus more accurately on key assessment areas. Cases (b) and (d) show higher accuracy, with the 99th percentile of the squared pixel error remaining below 0.03. Since the error maps display squared differences, this value corresponds to an absolute pixel-value difference of approximately 0.17 at the 99th percentile. Since the 45° ply-angle validation set verifies the ability to generalize across plies and the in-plane validation set verifies the ability to predict damage, the results confirm that the surrogate model not only exhibits cross-ply generalization capabilities but also achieves different damage severity levels prediction in in-plane testing, even under constrained input information (Table 3).

The metrics in Figure 8 show that the surrogate model preserves both pixel-level accuracy and perceptual consistency in intramaterial damage prediction. The RMSE values across the four validation sets remain concentrated near 0, indicating that most predicted pixels stay close to the ground-truth damage index. This behavior matters for BVID-like low-intensity damage regions because small numerical errors can determine whether a weak damage zone is retained or missed.

The LPIPS values also fluctuate near 0, which means that the predicted damage fields remain visually close to the FE results. The SSIM values stay above 0.7 and reach 1.0 in some cases, showing that the model preserves the spatial structure of the ground truth. Together, these metrics indicate that the model captures the main damage morphology and retains weak local features in low-intensity damage regions.

As illustrated in Figure 9, the DIR model enters a steady convergence regime within 300 training epochs, with the validation loss closely tracking the training loss throughout the process. The converged loss remains on the order of 10^{-4} , indicating stable reconstruction performance and good consistency between training and validation. It should be noted that the DIR and PIP branches do not share the same optimization schedule. In the present training setup, DIR is trained for 300 epochs with a learning-rate decay at epoch 200, so the corresponding transition appears earlier and remains relatively smooth in Figure 9(a) and (b). By contrast, PIP is trained for 1000 epochs with a learning-rate decay at epoch 500, and this schedule

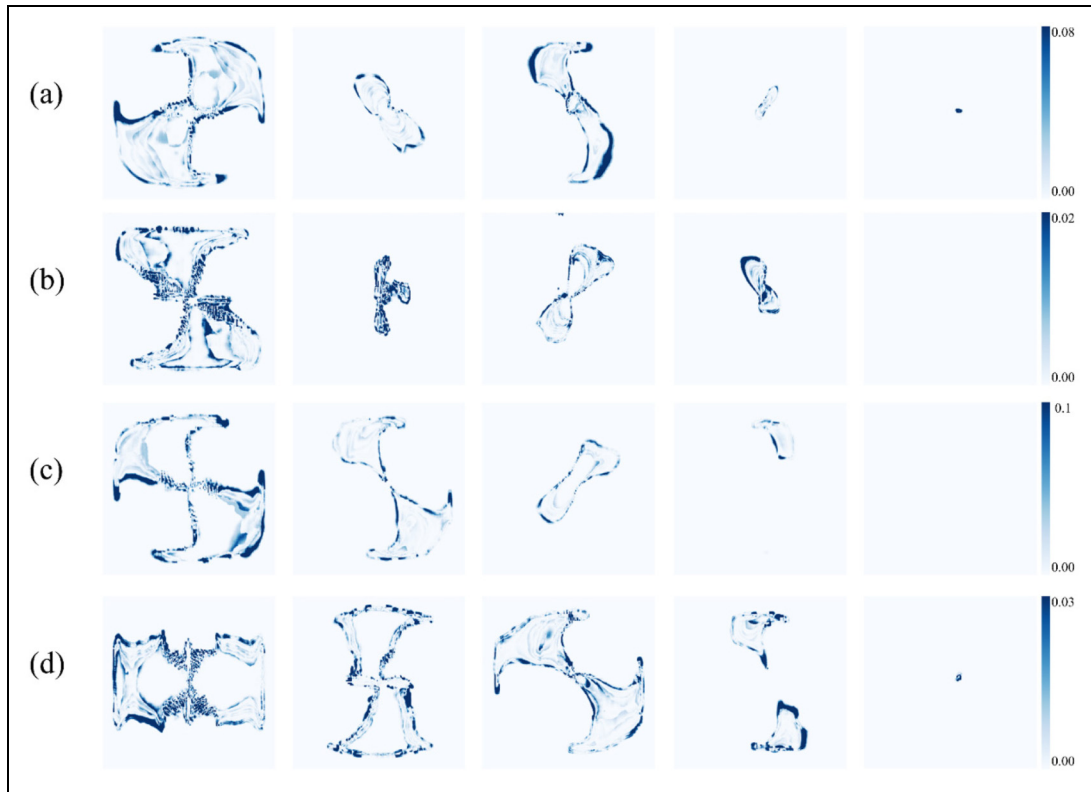


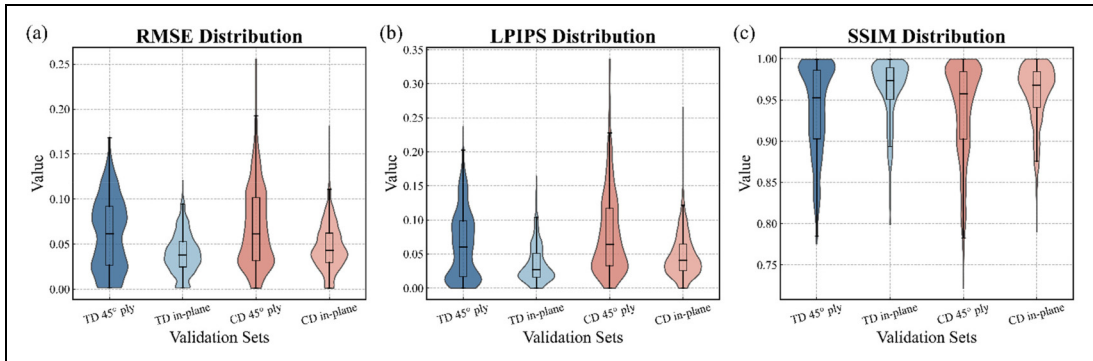
Figure 7. Error maps corresponding to the four validation sets in Figure 6. Panels (a)–(d) follow the same case and column order as Figure 6(a)–(d), respectively.

change produces the more visible bend observed in Figure 9(c) and (d) over the 500 to 600 epoch range. Therefore, the apparent jump in the PIP curves reflects a scheduled optimization transition rather than an instability of the prediction model. The final validation loss for the in-plane validation sets shows only minor order-of-magnitude discrepancies compared to training loss, which demonstrates the model's robust generalization. Although the cross-ply validation loss increases relative to the in-plane validation, the difference remains within less than an order of magnitude, confirming that the model maintains representational capacity for cross-ply damage prediction tasks.

The predictive capability in cross-ply scenarios derives from the characterization of fiber-induced anisotropy. By integrating the ply angle into the feature fusion process, the material-aware surrogate model captures the directional rotation of damage. Consequently, the DIR module reconstructs high-gradient damage zones that align with the principal fiber directions of each layer, demonstrating that the network identifies the anisotropic energy dissipation patterns inherent in laminated structures. Beyond pixel-level accuracy, the model captures physical symmetries governed by impact dynamics. It represents these nonlinear responses by mapping velocity vector components to the spatial distribution of the damage field. Furthermore, when the impact angle deviates from the normal direction, the PIP module captures the loss of center-symmetry in damage morphology, which is consistent with the change in the balance between the through-thickness normal momentum component and the in-plane tangential momentum components.

Table 3. Physical parameters of the representative cases shown in Figure 6.

Panel	Column	X	Y	Target layer	Fiber orientation	Lay-up
(a)	1	10°	10°	2	−45°	(±45°)2s
	2	70°	20°	7	−45°	(±45°)2s
	3	30°	60°	3	+45°	(±45°)2s
	4	40°	80°	6	+45°	(±45°)2s
	5	80°	80°	7	−45°	(±45°)2s
(b)	1	10°	10°	1	+15°	(±15°)2s
	2	30°	30°	4	90°	(90°)8
	3	20°	50°	8	+60°	(±60°)2s
	4	10°	80°	7	−75°	(±75°)2s
	5	70°	80°	4	−75°	(±75°)2s
(c)	1	10°	20°	8	+45°	(±45°)2s
	2	70°	40°	8	+45°	(±45°)2s
	3	40°	70°	1	+45°	(±45°)2s
	4	40°	80°	7	−45°	(±45°)2s
	5	80°	50°	6	+45°	(±45°)2s
(d)	1	10°	10°	6	90°	(90°)8
	2	10°	30°	6	+15°	(±15°)2s
	3	50°	60°	8	+60°	(±60°)2s
	4	60°	70°	8	+30°	(±30°)2s
	5	70°	80°	4	−75°	(±75°)2s

**Figure 8.** Violin plots for four validation sets under intramaterial prediction. (a) Root mean square error (RMSE) distribution; (b) learned perceptual image patch similarity (LPIPS) distribution; and (c) structural similarity index (SSIM) distribution.

Ablation study on intramaterial prediction. This subsection employs a modular ablation study to validate the predictive capability enhancement contributed by the CIE within the PIP architecture. Under a strictly controlled experimental framework, where network hyperparameters, training strategies, and input data consistency are rigorously preserved, the study conducts performance comparisons between models lacking the CIE module as a baseline and the full version.

As shown in Figure 10, cases I–IV denote four representative validation examples selected for qualitative ablation comparison. Figure 10(a) collects examples with relatively severe damage, whereas Figure 10(b)

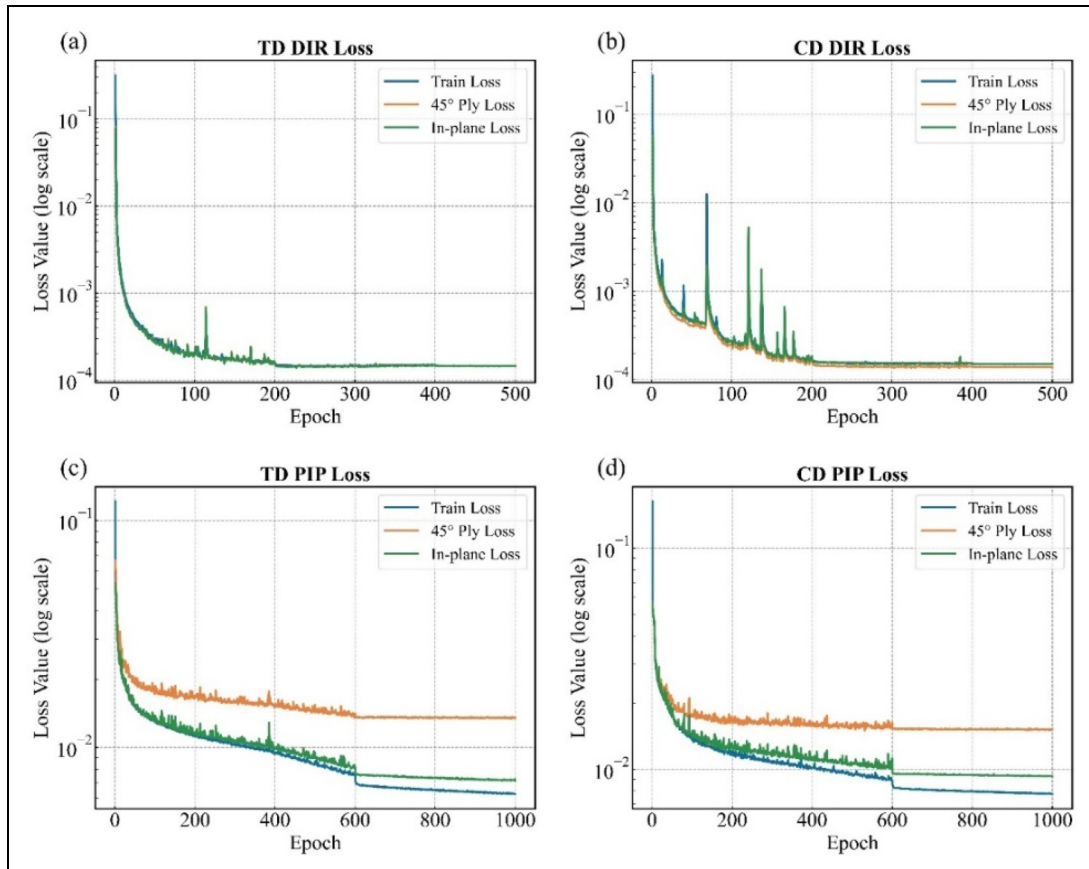


Figure 9. Loss function curves for the training and validation phases. (a) Tensile damage-damage image reconstructor (TD-DIR) loss; (b) compressive damage-damage image reconstructor (CD-DIR) loss; (c) TD-PIP loss; and (d) CD-PIP loss.

collects examples with relatively slight damage. Within each subfigure, the cases are arranged from more localized damage to more developed damage so that the comparison covers different damage levels. Under this ordering, the model without the CIE module tends to blur damage boundaries and lose morphology details, whereas the CIE-enhanced model preserves the damage shape and spatial extent more faithfully across both severe and slight damage cases. The corresponding physical parameters of the representative cases in Figure 10 are summarized in Appendix Tables A3 and A4.

In Tables 4 and 5, all the evaluation metrics present significant enhancement, with the LPIPS index demonstrating the most prominent optimization effect with the highest decrease of 56.5%, which indicates that the improved model mitigates the image blurring and pixel-point discontinuity problems that existed without the introduction of the CIE network at the level of visual perception, a conclusion that is in line with the phenomenon observed in the qualitative analysis. Second, the highest decrease of 28.7% in the RMSE index verifies the improvement of the prediction accuracy at the pixel level, confirming that the introduction of the CIE network can achieve more accurate numerical prediction of the damage region while ensuring the perception quality. The highest increase of 4.9% in the SSIM index further corroborates the utility of the CIE to enhance the expressive power of the network in terms of structural similarity.

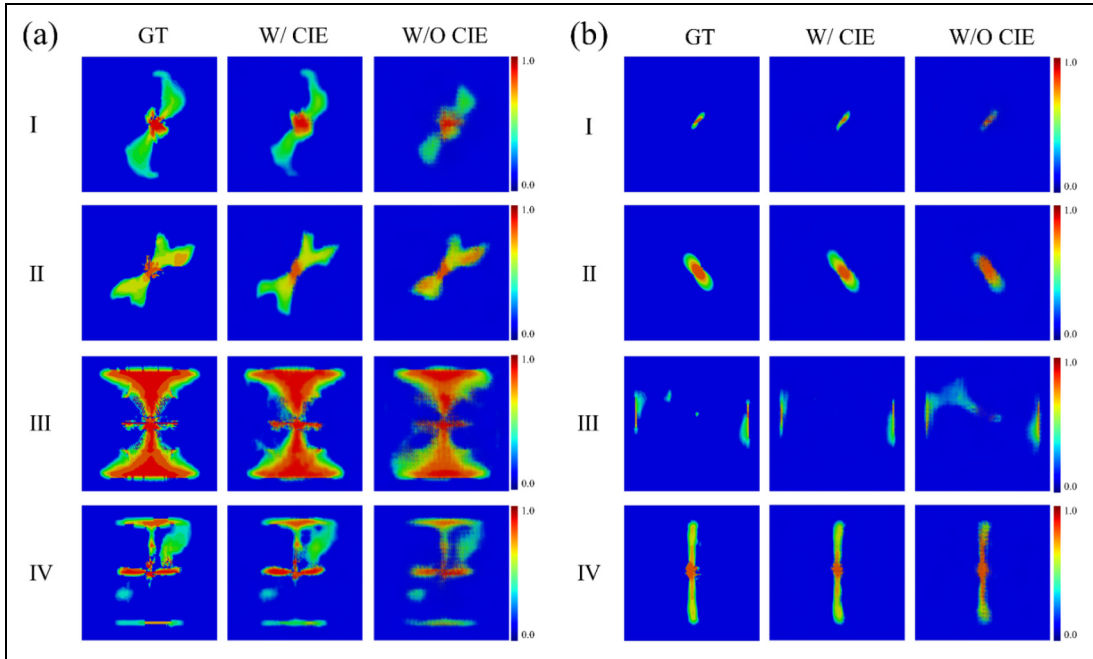


Figure 10. Comparative ablation analysis on single-material damage prediction in laminated composite. Cases I–IV denote four representative validation examples selected from the validation set. (a) Relatively severe damage cases. (b) Relatively slight damage cases. Within each subfigure, the cases are ordered from more localized damage to more developed damage. The three columns correspond to ground truth, prediction with CIE, and prediction without CIE.

Table 4. Performance comparison of baseline versus CIE-enhanced models on the tensile damage (TD) validation set.

Method	TD in-plane			TD 45° ply		
	RMSE	LPIPS	SSIM	RMSE	LPIPS	SSIM
Without CIE	0.0557	0.0806	0.9184	0.0683	0.1011	0.8987
With CIE	0.0397	0.0351	0.9630	0.0612	0.0636	0.9390
Numerical change (%)	28.7↓	56.5↓	4.9↑	10.4↓	37.1↓	4.5↑

RMSE: root mean square error; LPIPS: learned perceptual image patch similarity; SSIM: structural similarity index.

The network architecture introducing the CIE module overcomes the above limitations. This is achieved because the PIP module essentially performs the task of converting low-dimensional features to high-dimensional features. Due to the huge dimensional gap between the low-dimensional input and the target high-dimensional space in this task, it is difficult to achieve effective feature enhancement by simply relying on an MLP. Experiments demonstrate that this simple mapping not only introduces significant computational overhead but also delivers mapping quality insufficient to meet accuracy requirements. More importantly, considering that the high-dimensional feature map still needs to characterize the spatial distribution of the impact damage field, we combine the local feature integration advantage of CNNs with the global mapping capability of MLP: the local information integration capability of CNNs can capture the morphological

Table 5. Performance comparison of baseline versus CIE-enhanced models on the compressive damage (CD) validation set.

Method	CD in-plane			CD 45° ply		
	RMSE	LPIPS	SSIM	RMSE	LPIPS	SSIM
Without CIE	0.0564	0.0843	0.9256	0.0720	0.1055	0.9142
With CIE	0.0466	0.0478	0.9576	0.0695	0.0799	0.9373
Numerical change (%)	17.4↓	43.3↓	3.5↑	3.5↓	24.3↓	2.5↑

RMSE: root mean square error; LPIPS: learned perceptual image patch similarity; SSIM: structural similarity index.

Table 6. Evaluation metrics performance of intermaterial models on four material validation sets.

Material	Tensile damage			Compressive damage		
	RMSE	LPIPS	SSIM	RMSE	LPIPS	SSIM
T700GC/M2I	0.0431	0.0384	0.9627	0.0481	0.0478	0.9572
AS4/8552	0.0437	0.0356	0.9587	0.0502	0.0482	0.9546
CFRP	0.0462	0.0414	0.9540	0.0393	0.0352	0.9682
T700/790I	0.0416	0.0325	0.9625	0.0411	0.0394	0.9655

RMSE: root mean square error; LPIPS: learned perceptual image patch similarity; SSIM: structural similarity index.
Bold values indicate the best performance for each metric among the four material validation sets.

features of the damage field, while MLP is responsible for realizing the cross-dimensional mapping. This coupling architecture ultimately achieves a significant improvement in damage prediction accuracy through synergistic optimization in different directions.

This ablation study confirms that the CIE plays a critical role in nonlinear mapping, compensating for the limitations of pure MLP architectures in predicting LVI damage fields under complex impact conditions. By resolving the feature representation bottleneck in low-dimensional to high-dimensional nonlinear mapping, we establish a reliable algorithmic foundation for damage field prediction under complex intermaterial scenarios.

Intermaterial damage field prediction via the material-aware surrogate model

Intermaterial results. The intermaterial evaluation used the same model architecture, preprocessing steps, and metric definitions for all materials. The three additional material systems were evaluated with a fixed $\pm 30^\circ$ lay-up, allowing the comparison to focus on material-dependent changes in damage intensity under the same stacking condition.

As shown in Table 6, the intermaterial model shows competitive performance relative to the intramaterial results reported in Tables 4 and 5. In the TD damage prediction task, the metrics are close to those of the intramaterial model. In the CD damage prediction task, the intermaterial model gives locally lower RMSE values, while the other metrics remain comparable. The two validation settings are not identical in their data distributions, so this comparison should be read as evidence of material-dependent intensity adaptation rather than a direct claim of higher generalization difficulty. In the intermaterial prediction, the best validation metrics do not always correspond to the primary material class in the dataset. This pattern is consistent with the asymmetric data distribution used during model development and suggests that feature fusion helps the model respond to material type variations.

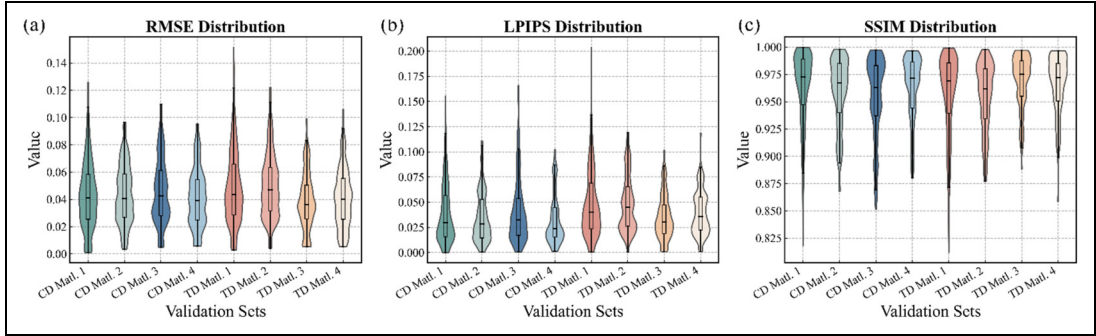


Figure 11. Violin plots for eight validation sets under intermaterial prediction. (a) Root mean square error (RMSE) distribution; (b) learned perceptual image patch similarity (LPIPS) distribution; (c) structural similarity index (SSIM) distribution. Matl. 1 for T700GC/M21, Matl. 2 for AS4/8552, Matl. 3 for CFRP, Matl. 4 for T700/7901.

As shown in Figure 11, the validation metrics are concentrated in the desired range for all four materials. T700GC/M21 has the largest training subset, so it would be natural to expect the best validation scores on this material. The results do not always follow that pattern. This suggests that the model learns two related effects: shared damage-field structure across materials and material-dependent intensity changes introduced through feature fusion. Because the three additional material systems use a fixed $\pm 30^\circ$ lay-up, this intermaterial study mainly evaluates material-dependent intensity adaptation under a controlled lay-up setting rather than full cross-material and cross-stacking generalization.

As discussed above, material type is not the primary factor controlling the global damage skeleton in the representative cases, but it affects damage intensity and threshold. The case-level visualizations in Appendix Figures A1–A4 therefore remain close to those in the intramaterial section. The intermaterial analysis focuses on RMSE, LPIPS, and SSIM because these metrics are more sensitive to material-specific intensity changes on a similar damage skeleton, including weak low-intensity regions. Broader cross-material and cross-stacking generalization will require material-wise lay-up coverage beyond the present fixed $\pm 30^\circ$ subsets.

For Table 6 and Figure 11, the T700GC/M21 validation subset spans seven lay-up families: $(0^\circ)_8$, $(\pm 15^\circ)_{2s}$, $(\pm 30^\circ)_{2s}$, $(\pm 45^\circ)_{2s}$, $(\pm 60^\circ)_{2s}$, $(\pm 75^\circ)_{2s}$, and $(90^\circ)_8$. The AS4/8552, CFRP, and T700/7901 subsets each use the fixed $(\pm 30^\circ)_{2s}$ lay-up. Table 6 reports material-wise averages over these validation subsets, while Figure 11 shows the corresponding per-image metric distributions. Comparisons among the three additional materials are therefore made under a matched lay-up condition. Comparisons with T700GC/M21 also reflect its broader lay-up distribution.

The observed results can be explained by two factors. First, the in-plane validation sets in the intermaterial experiments give the model a focused view of damage morphology. Second, the material-aware feature fusion uses the material descriptor to adjust the learned response for different material thresholds. This helps the model recognize material-specific intensity changes without treating every material system as a separate prediction problem.

This sensitivity is also visible in the spatial consistency across material systems. The material-aware surrogate model separates the geometric shape of damage from its numerical intensity. The layup controls the global damage pattern, while material-specific properties shift the damage threshold. The feature fusion mechanism therefore allows damage area and intensity to change with material properties while preserving the main topological structure. This distinction matters for BVID-like low-intensity damage regions because weak damage zones may share the same shape but differ in pixel intensity across materials. This

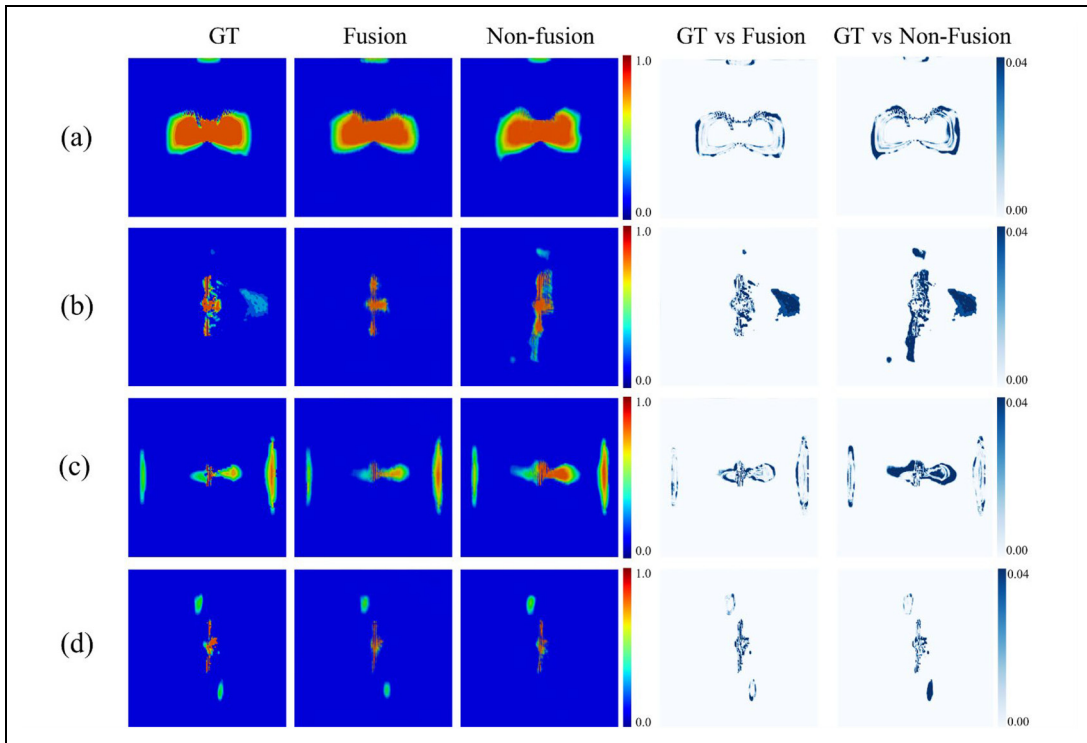


Figure 12. Comparison of predicted damage fields for four representative examples from the dominant T700GC/M21 subset in the intermaterial ablation study. The panels illustrate typical fusion and nonfusion error modes; material-wise performance across all four material systems is summarized in Figure 11 and the material-wise metric table. Cases (a)–(d) are selected from the validation set to cover damage patterns ranging from localized regions to larger and more developed regions. Columns 1–3 show ground truth (GT), fusion method results, and nonfusion method results. Columns 4 and 5 show the error maps between GT and the fusion or nonfusion predictions.

interpretation is consistent with cross-material transfer studies in which material features are used to separate shared response trends from material-specific shifts (Zhang and Tang, 2025; Zhang et al., 2025a).

Ablation study on intermaterial prediction. This ablation study tests whether feature fusion reduces prediction error in the controlled intermaterial setting. Because the three additional materials are evaluated under a fixed $\pm 30^\circ$ lay-up, the comparison focuses on material-dependent intensity adaptation, especially near the damage threshold where BVID-like low-intensity regions can disappear if material-specific threshold information is not encoded.

As shown in the first three columns of Figure 12, the nonfusion model produces several visible error modes during intermaterial validation. In case (a), it captures part of the damaged area but distorts nearby boundaries, which suggests weak modeling of damage propagation outside the main impact zone. Case (b) includes false positives in undamaged areas, while case (c) overestimates high-intensity damage. Cases (b) and (d) also miss weak damage regions. Together, these errors show that the nonfusion model mixes material-related intensity changes with the global damage pattern. With feature fusion, the material-aware surrogate model adjusts damage intensity using the material descriptor in the input vector and retains low-intensity transition regions near the damage threshold.

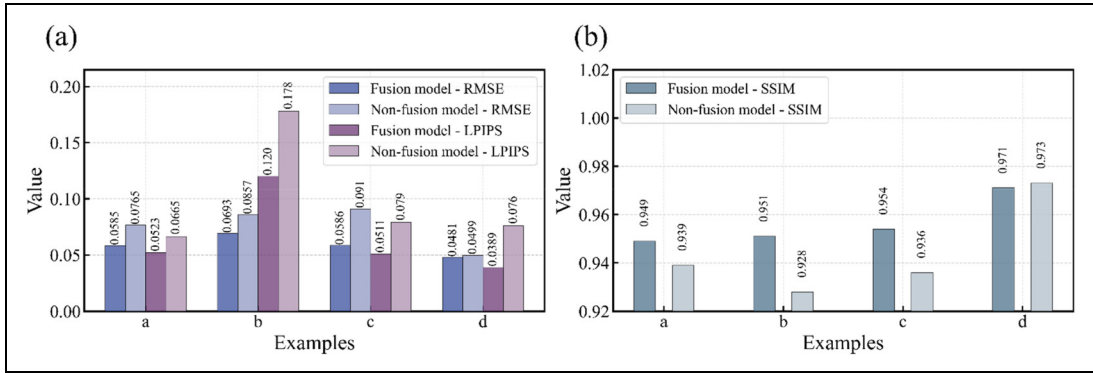


Figure 13. Comparison plots of the four sample performance metrics. (a) Comparison plot of root mean square error (RMSE) and learned perceptual image patch similarity (LPIPS); and (b) comparison plot of structural similarity index (SSIM).

As shown in the first three columns of Figure 12, cases (a)–(d) are representative examples from the dominant T700GC/M21 subset. They are used here to show the main error modes of the fusion and nonfusion architectures, not to claim that these four panels cover all material systems. The cases span from localized to more developed damage patterns, so the comparison still covers different levels of spatial complexity within this subset. Under this setting, the nonfusion architecture shows distorted damage boundaries, false prediction of undamaged regions, overestimated damage extent, and omission of minor damage regions. The fusion architecture gives predictions closer to the ground truth for these examples, which indicates better encoding of material-sensitive damage morphology and spatial organization. Material-wise quantitative evidence for the intermaterial setting is provided in Figure 11 and the material-wise metric table, and the physical parameters of Figure 12 examples are listed in Appendix Table A5.

The error maps in the last two columns of Figure 12 show the same trend for these selected examples. The nonfusion model produces larger dark-blue areas, with pixel errors close to 0.04, mainly around transitions between damaged and undamaged regions. These zones are where weak damage features are most easily lost. The fused model reduces these errors, indicating that material-aware feature fusion helps preserve fine damage boundaries and low-intensity morphology during cross-material prediction. Similar sensitivity to localized low-intensity impact damage has been reported in recent LVI characterization and damage-tolerance studies (Koçak et al., 2026; Yue et al., 2026; Zhao et al., 2026).

As shown in Figure 13, the fusion architecture performs better than the nonfusion architecture in cases (a)–(c) across the reported metrics. Case (d) is less clear: the SSIM difference is only 0.002, which falls within small statistical variation and should be treated as comparable performance rather than a meaningful gap. Even in this case, the fusion model still has lower RMSE and LPIPS.

The nonfusion errors mainly arise because, under a controlled lay-up, damage shapes can look similar across material systems while intensity levels shift with material thresholds. Without feature interaction, the DIR module tends to quantize visually similar inputs into nearly identical latent features. This limits the PIP module’s ability to learn small material-specific differences. Feature fusion reduces this ambiguity by giving DIR a material-discriminative latent representation even when the damage fields look alike. During feature encoding, the fused architecture links material type with damage-field morphology, allowing PIP to learn nonlinear mappings that depend on both the material parameter and the damage pattern.

These results explain the role of feature fusion in the intermaterial setting. It separates latent features across material systems and reduces errors in weakly damaged regions under the fixed-lay-up comparison.

The material-aware surrogate model therefore preserves low-intensity damage morphology more consistently when damage thresholds vary by material. Broader cross-material and cross-stacking generalization will require additional material-wise lay-up coverage.

Concluding remarks

This study proposes a material-aware surrogate model for predicting LVI damage in laminated composites under complex impact conditions. The main contribution is a feature fusion mechanism that links internal material features with external impact dynamics so that damage morphology and material-specific damage thresholds can be learned together. The framework is evaluated through intramaterial generalization and intermaterial adaptability tests. The main findings are summarized as follows:

1. *Accurate and rapid damage prediction.* The material-aware surrogate model evaluates damage morphology with a 5 ms inference time. It preserves BVID-like low-intensity regions in FE-generated fiber damage fields and provides an efficient alternative to repeated numerical simulations.
2. *Material-aware adaptability under controlled lay-up conditions.* By integrating material descriptors with impact and laminate parameters, the framework adapts damage intensity across material systems under the present fixed-lay-up intermaterial setting. Broader cross-material and cross-stacking generalization requires additional material-wise lay-up coverage.
3. *Analysis of fiber-damage response characteristics.* The model shows that the external impact direction governs the symmetry of the fiber-damage distribution, the fiber orientation induces morphological rotation, and the material type mainly shifts the damage threshold while preserving the global pattern.

Minor damage morphology remains difficult to characterize across material systems, but the proposed framework predicts LVI fiber-damage fields with consistent accuracy under the tested impact conditions. Future developments will need to first extend the same framework to additional damage targets, especially matrix TD and matrix CD, when corresponding datasets are available. We will also need to incorporate physics-informed mechanisms to improve the mechanical consistency of the predicted damage fields. Building on this step, time-series deep learning architectures can be used to move from static damage assessment to temporal damage evolution tracking.

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Data availability

The code and dataset have been open-sourced at: <https://github.com/dazhizhao/CLiDPN>. The data will also be made available by the corresponding author upon reasonable request.

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Appendix

A.1 Material properties and FE setup

The FE simulations use the material parameters listed in Table A1. These same parameters define the damage-field images used for training and validation, so the results in the main text and Appendix come from the same FE data source.

The eight-ply laminate uses surface-to-surface contact between adjacent plies. Algorithm A1 gives the Abaqus Explicit contact-definition procedure used for those interfaces.

Algorithm A1. Automated interlaminar contact definition for the 8-ply Abaqus Explicit model.

Algorithm 1 Automated Interlaminar Contact Definition for an 8-ply Laminate

Require: Laminate part P , ply thickness $t = 0.25$ mm, total plies $N = 8$

Require: Impactor surface S_{imp} , support surface S_{sup}

- 1: Generate the solid laminate mesh from the plate surface and divide it into N ply groups
 - 2: **for** $i = 1$ to N **do**
 - 3: Create the top surface T_i of ply i from the element bounding box of ply i
 - 4: Create the bottom surface B_i of ply i from the element bounding box of ply i
 - 5: **end for**
 - 6: Define a single interaction property IntProp:
 - normal behavior = hard contact with separation allowed
 - tangential behavior = penalty friction with coefficient $\mu = 0.3$
 - 7: **for** $i = 1$ to $N - 1$ **do**
 - 8: Assign a finite-sliding surface-to-surface contact pair between B_i and T_{i+1}
 - 9: Use kinematic contact enforcement and the shared property IntProp
 - 10: **end for**
 - 11: Assign the impactor-to-laminate contact pair between S_{imp} and T_1
 - 12: Assign the laminate-to-support contact pair between B_N and S_{sup}
 - 13: Solve the impact event in Abaqus/Explicit with the above contact definitions
-

Table A1. Material properties.

Material	ρ (10 ³ kg/ m ³)	E_1 (GPa)	$E_2 = E_3$ (GPa)	$G_{12} = G_{13}$ (MPa)	$n_{12} = n_{13}$	n_{23}	X_t (MPa)	X_c (MPa)	Y_t (MPa)	Y_c (MPa)	F_{3t} (MPa)	F_{3c} (MPa)	$f_{12} = f_{13}$ (MPa)	f_{13} (MPa)	G_{ft} (kJ/ m ²)	G_{fc} (kJ/ m ²)	G_{mt} (kJ/ m ²)	G_{mc} (kJ/ m ²)
T700GC/ M21 (Li et al., 2024)	1.60	130	7.7	4800	0.33	0.35	2080	1250	60	140	290	290	110	110	133	40	0.6	1.48
ASA/8552 (Falcó et al., 2022)	1.58	137.1	8.8	4900	0.31	0.49	2106	1675	74	322	74	322	110	110	125	61	0.30	1.54
CFRP (Shi et al., 2012)	1.60	153	10.3	6000	0.30	0.40	2537	1580	82	236	82	236	90	40	91.6	79.9	0.22	1.6
T700/7901 (Hou et al., 2024)	1.72	115	9.83	5380	0.31	0.44	2300	1050	50	100	50	100	55	55	40	20	0.6	1.72

A.2 Intermaterial validation examples

Figures A1–A4 show intermaterial validation cases for fiber tensile and fiber compressive damage. They give the image-level counterpart to the metric distributions in the main text: predicted fields and pixel-wise error maps for all four material systems. See Table A2.

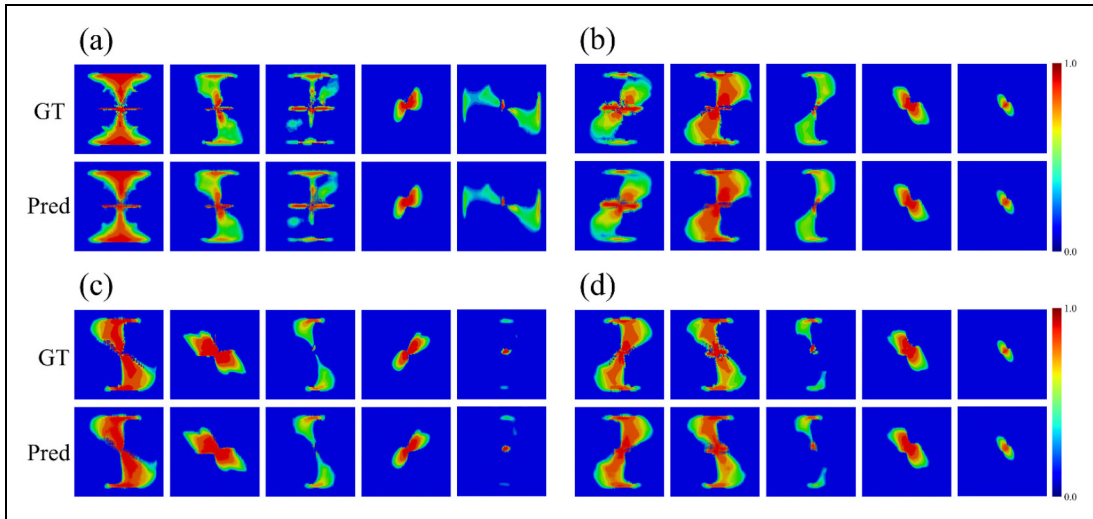


Figure A1. Validation set comparisons between ground truth (GT) and predicted damage fields (Pred) in intermaterial experiments under TD. (a) T700GC/M21; (b) AS4/8552; (c) CFRP; and (d) T700/7901.

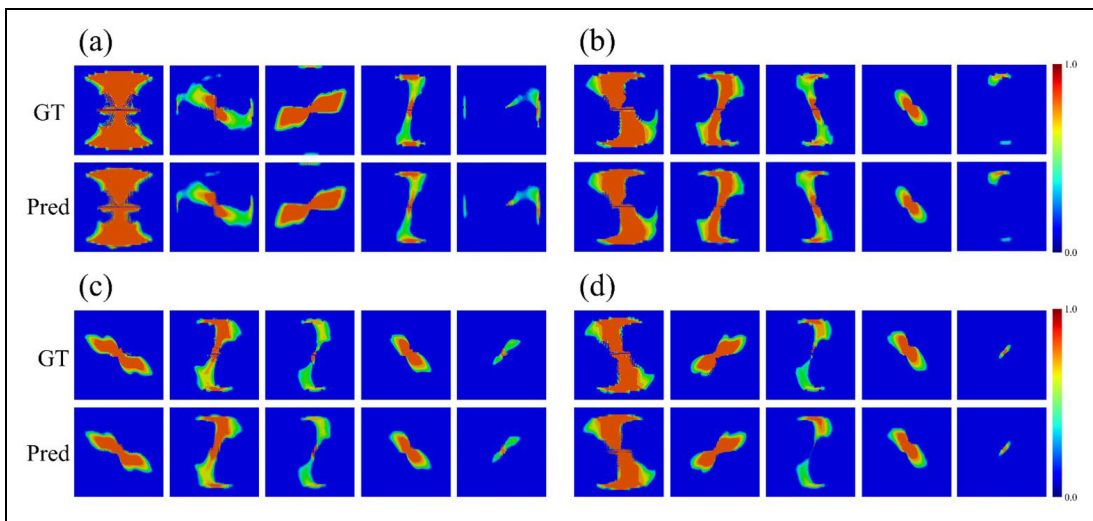


Figure A2. Validation set comparisons between ground truth (GT) and predicted damage fields (Pred) in intermaterial experiments under CD. (a) T700GC/M21; (b) AS4/8552; (c) CFRP; and (d) T700/7901.

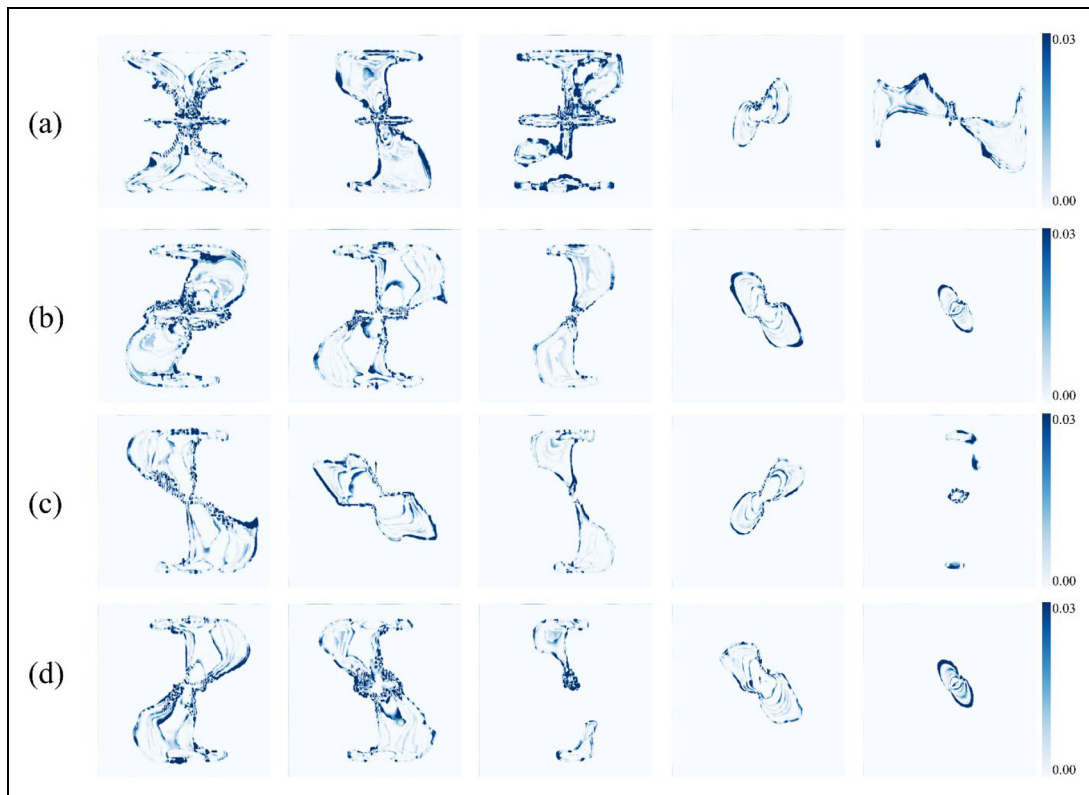


Figure A3. Error map on four types of material under TD. (a) T700GC/M21; (b) AS4/8552; (c) CFRP; and (d) T700/7901.

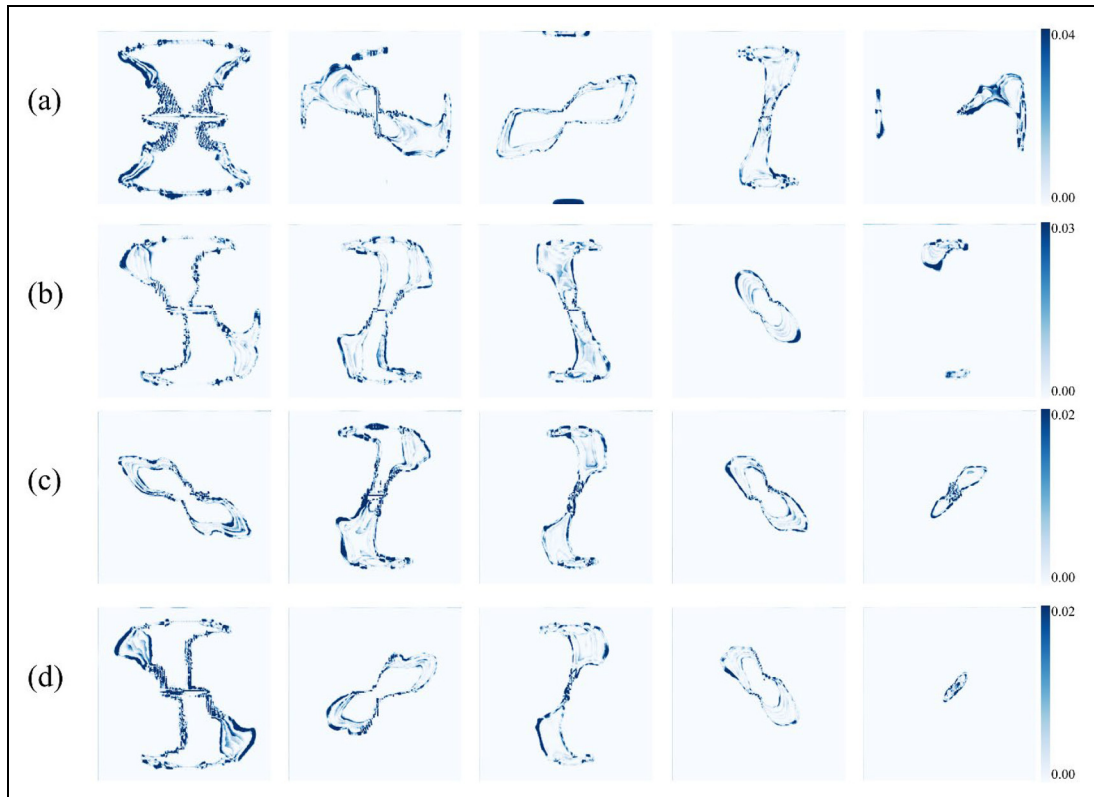


Figure A4. Error map on four types of material under CD. (a) T700GC/M21; (b) AS4/8552; (c) CFRP; and (d) T700/7901.

Table A2. Physical parameters of the representative cases shown in Figures A1–A4.

Damage type	Panel	Material system	Column	X	Y	Target layer	Fiber orientation	Lay-up
Fiber tensile	(a)	T700GC/M2I	1	20°	10°	2	0°	(0°)8
			2	50°	10°	3	+15°	(±15°)2s
			3	20°	20°	4	−15°	(±15°)2s
			4	20°	70°	8	+15°	(±15°)2s
			5	70°	20°	1	+75°	(±75°)2s
	(b)	AS4/8552	1	20°	10°	4	−30°	(±30°)2s
			2	40°	10°	2	−30°	(±30°)2s
			3	70°	20°	2	−30°	(±30°)2s
			4	50°	60°	7	−30°	(±30°)2s
			5	50°	80°	7	−30°	(±30°)2s
	(c)	CFRP	1	50°	10°	1	+30°	(±30°)2s
			2	10°	30°	7	−30°	(±30°)2s
			3	70°	50°	1	+30°	(±30°)2s
			4	50°	60°	8	+30°	(±30°)2s
			5	60°	60°	4	−30°	(±30°)2s
	(d)	T700/790I	1	40°	20°	2	−30°	(±30°)2s
			2	20°	30°	3	+30°	(±30°)2s
			3	20°	70°	3	+30°	(±30°)2s
			4	20°	60°	7	−30°	(±30°)2s
			5	50°	80°	7	−30°	(±30°)2s
Fiber compressive	(a)	T700GC/M2I	1	10°	10°	8	0°	(0°)8
			2	30°	30°	6	+60°	(±60°)2s
			3	10°	20°	1	+15°	(±15°)2s
			4	70°	40°	7	−15°	(±15°)2s
			5	60°	60°	7	−75°	(±75°)2s
	(b)	AS4/8552	1	40°	10°	8	+30°	(±30°)2s
			2	60°	20°	7	−30°	(±30°)2s
			3	60°	30°	6	+30°	(±30°)2s
			4	70°	60°	2	−30°	(±30°)2s
			5	10°	80°	6	+30°	(±30°)2s
	(c)	CFRP	1	10°	10°	2	−30°	(±30°)2s
			2	20°	50°	7	−30°	(±30°)2s
			3	70°	40°	7	−30°	(±30°)2s
			4	60°	40°	2	−30°	(±30°)2s
			5	50°	60°	3	+30°	(±30°)2s
	(d)	T700/790I	1	10°	10°	8	+30°	(±30°)2s
			2	40°	10°	1	+30°	(±30°)2s
			3	30°	70°	7	−30°	(±30°)2s
			4	20°	70°	2	−30°	(±30°)2s
			5	50°	80°	3	+30°	(±30°)2s

The fiber tensile damage cases correspond to Figures A1 and A3, while the fiber compressive damage cases correspond to Figures A2 and A4. Figures A3 and A4 show the error maps of the cases presented in Figures A1 and A2, respectively.

A.3 Representative cases used in ablation comparisons

Tables A3 and A4 list representative fiber tensile and fiber compressive damage cases used in the single-material ablation comparisons in Figure 10. The cases were selected independently from their respective validation sets to cover different damage levels and morphologies. Labels I–IV indicate the display order within each damage task and do not denote paired physical conditions across the two tables. For each case, the ground truth, CIE-enhanced prediction, and baseline prediction use the same X , Y , N , and θ . Table A5 lists the representative cases used for the intermaterial ablation visualization in Figure 12. Here, N denotes the sequence number of the target ply through the laminate thickness, and θ denotes the fiber orientation angle of that ply.

Table A3. Candidate physical parameters of the representative cases in Figure 10 screened from fiber tensile damage.

Panel	Material system	Y (deg)	X (deg)	Target ply N	Theta (deg)
(a)-I	T700GC/M21	10	60	4	−45
(a)-II	T700GC/M21	40	40	8	45
(a)-III	T700GC/M21	10	10	1	0
(a)-IV	T700GC/M21	30	10	4	−15
(b)-I	T700GC/M21	80	70	8	45
(b)-II	T700GC/M21	40	80	7	−45
(b)-III	T700GC/M21	70	60	1	75
(b)-IV	T700GC/M21	10	10	5	90

Table A4. Candidate physical parameters of the representative cases in Figure 10 screened from fiber compressive damage.

Panel	Material system	Y (deg)	X (deg)	Target ply N	Theta (deg)
(a)-I	T700GC/M21	20	60	3	60
(a)-II	T700GC/M21	20	20	3	45
(a)-III	T700GC/M21	10	20	6	0
(a)-IV	T700GC/M21	70	40	6	15
(b)-I	T700GC/M21	80	70	3	45
(b)-II	T700GC/M21	60	80	2	−45
(b)-III	T700GC/M21	80	10	8	75
(b)-IV	T700GC/M21	10	10	3	90

Table A5. Physical parameters of the dominant-subset representative cases in Figure 12.

Case	Material system	Y (deg)	X (deg)	Target ply N	Theta (deg)
(a)	T700GC/M21	50	10	1	0
(b)	T700GC/M21	10	20	4	90
(c)	T700GC/M21	40	60	6	90
(d)	T700GC/M21	20	20	4	−60

A.4 Validation protocol consistency

To isolate the effect of validation-set composition, the frozen T700GC/M21 intramaterial models were evaluated on both the full validation set and the non-45° angle-ply subset. The same checkpoints, preprocessing, and metric definitions were used in both evaluations. Figure A5 therefore shows how RMSE, LPIPS, and SSIM change when the evaluation is restricted to the non-45° subset. The differences are limited for most metrics, with a more visible change in SSIM for fiber compressive damage.

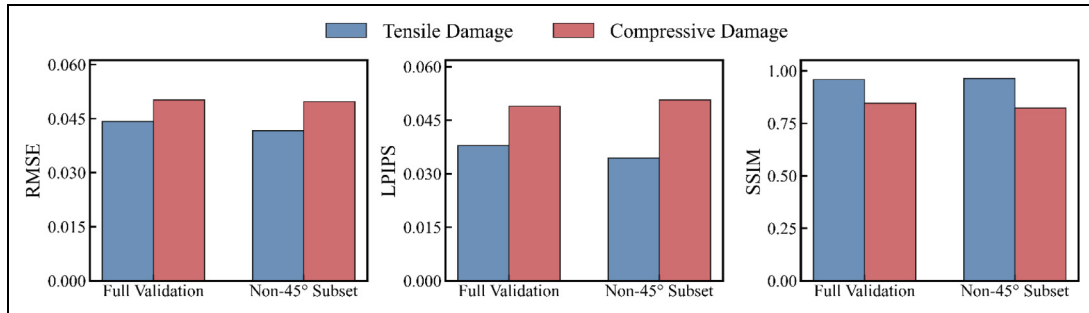


Figure A5. Effect of validation-set composition on the T700GC/M21 intramaterial models. The same frozen checkpoints are evaluated on the full validation set and the non-45° angle-ply subset. Bars report RMSE, LPIPS, and SSIM for fiber tensile damage and fiber compressive damage. RMSE: root mean square error; LPIPS: learned perceptual image patch similarity; SSIM: structural similarity index.

A.5 Intramaterial and intermaterial comparison for T700GC/M21

Figures A6 and A7 compare the intramaterial and intermaterial models on T700GC/M21 under the held-out $[+45/-45]_{2s}$ cross-ply case. This test asks whether the material-conditioned, cross-material model loses accuracy on the material with the largest training set.

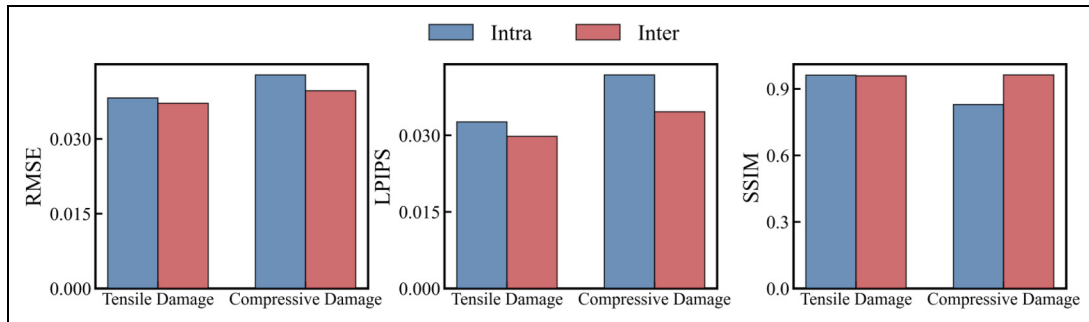


Figure A6. Intramaterial and intermaterial metrics for T700GC/M21. Quantitative comparison of intramaterial and intermaterial predictions for T700GC/M21 under the held-out $[+45/-45]_{2s}$ stacking sequence. For fiber tensile damage, the two models are nearly tied: the intermaterial model has slightly lower RMSE and LPIPS, with a small drop in SSIM. For fiber compressive damage, the intermaterial model improves all three metrics, most clearly SSIM, indicating better preservation of the damage shape. RMSE: root mean square error; LPIPS: learned perceptual image patch similarity; SSIM: structural similarity index.

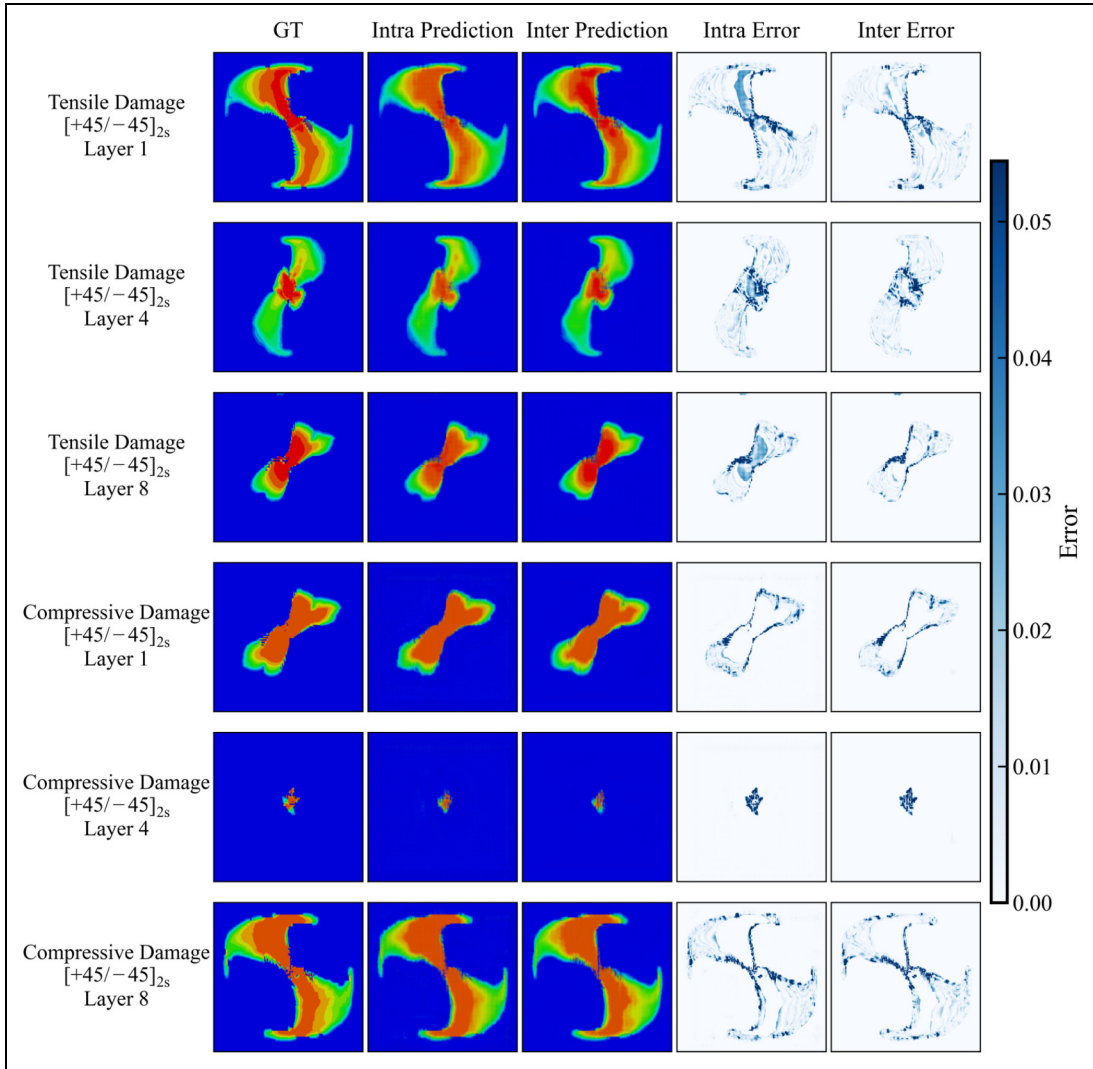


Figure A7. T700GC/M21 cross-ply prediction examples. Representative T700GC/M21 $[+45/-45]_{2s}$ damage fields compare ABAQUS ground truth with intramaterial and intermaterial predictions. Both models recover the main damage location and overall morphology. The intermaterial prediction remains close to the ground truth and gives error maps comparable to the intramaterial prediction, even though it is trained across material systems.

Both views give the same answer: the intermaterial model remains close to the intramaterial model. For fiber tensile damage, their metrics are nearly the same. For fiber compressive damage, the intermaterial model gives a higher SSIM, which means it preserves the main damage shape more faithfully while keeping similar error maps.

A.6 Unseen stacking-sequence generalization

Figures A8–A11 test a small set of stacking sequences that were not part of the original training design. The AS4/8552 cases use $[+15/-15]_{2s}$ and $[+45/-45]_{2s}$, while the T700GC/M21 cases use $[+10/-10]_{2s}$ and one quasi-isotropic $[45/90/-45/0]_s$ laminate. See Figures A9 and A10.

On AS4/8552, the predictions follow the ABAQUS fields for both unseen sequences, although fiber compressive damage remains harder because weak, sparse damage regions penalize small local mismatches. On T700GC/M21, both surrogate settings recover the main damage zones for the quasi-isotropic laminate in this additional unseen-sequence check. This case does not provide systematic quasi-isotropic coverage; it is a limited validation example outside the main Table 1 training dataset. The result suggests that the model has learned part of the relation between layer position, fiber orientation, and damage morphology, rather than only memorizing the original $[+x/-x]_{2s}$ family.

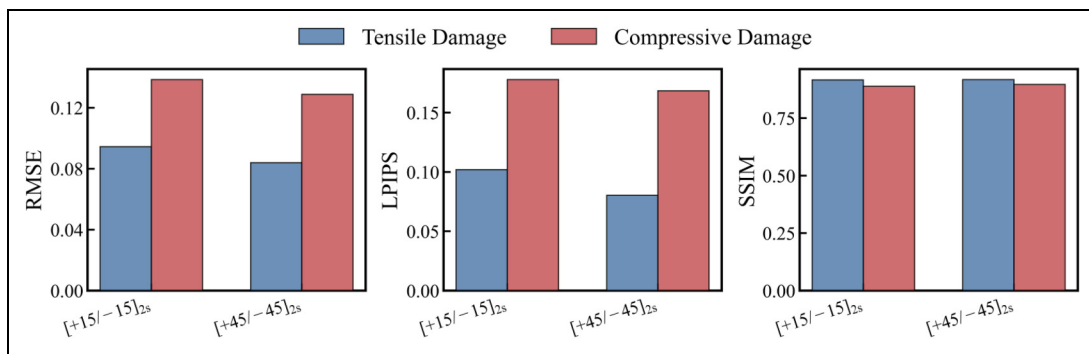


Figure A8. AS4/8552 unseen-sequence metrics. The intermaterial model is tested on AS4/8552 laminates with unseen but within-distribution stacking sequences, $[+15/-15]_{2s}$ and $[+45/-45]_{2s}$. Each bar averages RMSE, LPIPS, and SSIM over 64 impact positions and 8 layers. The model gives usable agreement with the ABAQUS fields for both sequences, with fiber compressive damage remaining harder to predict than fiber tensile damage. RMSE: root mean square error; LPIPS: learned perceptual image patch similarity; SSIM: structural similarity index.

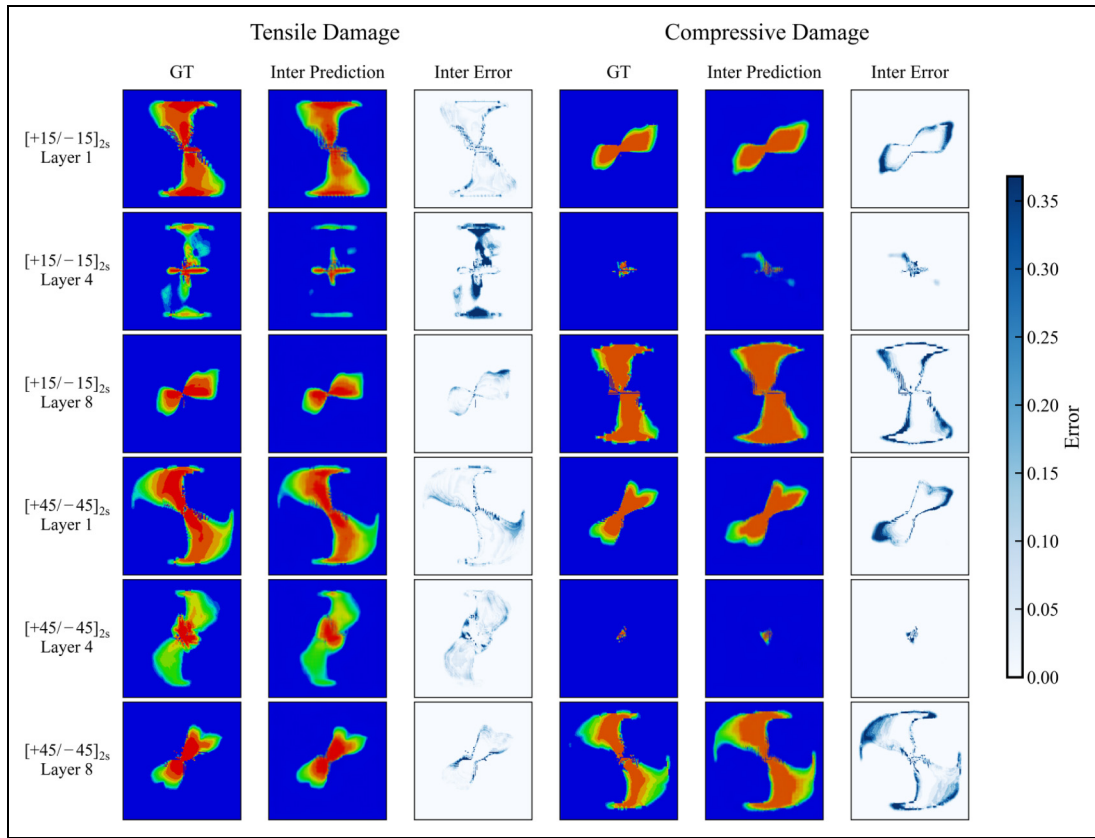


Figure A9. AS4/8552 unseen-sequence prediction examples. Representative AS4/8552 examples compare ABAQUS fields with intermaterial predictions for $[+15/-15]_{2s}$ and $[+45/-45]_{2s}$. The panels show ground truth, prediction, and pixel-wise error maps for selected layers. The predicted fields follow the dominant damage regions and the layer-dependent changes in shape, which provides the main visual check for generalization to these unseen lay-ups.

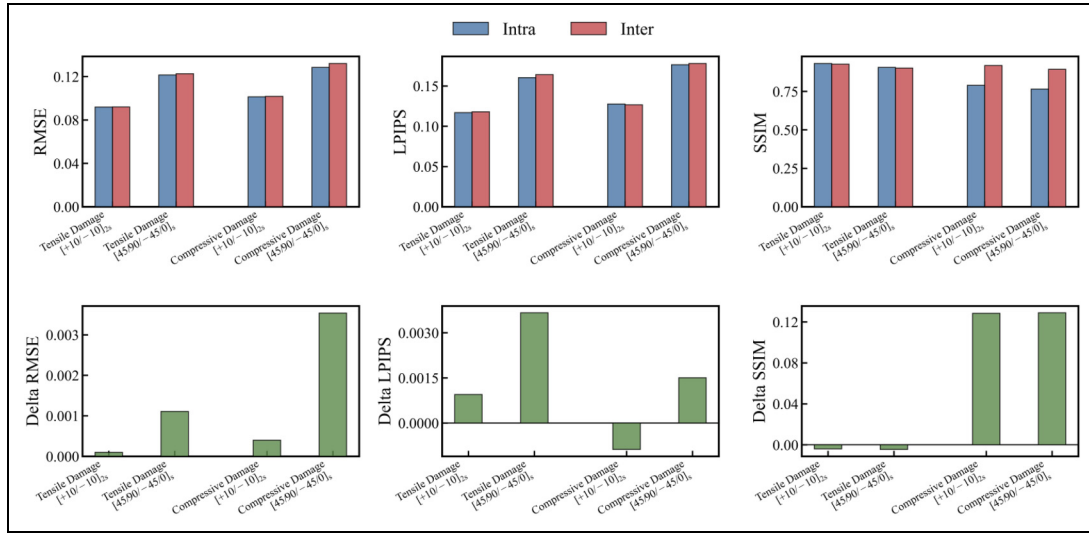


Figure A10. T700GC/M2I unseen-sequence metrics. Unseen T700GC/M2I stacking sequences [+10/-10]_{2s} and [45/90/-45/0]_s are evaluated with both intramaterial and intermaterial surrogates. The top row reports RMSE, LPIPS, and SSIM. The bottom row plots inter-minus-intra differences. RMSE and LPIPS remain close for the two models, while the positive SSIM differences for fiber compressive damage show a clearer structural advantage for the intermaterial model. RMSE: root mean square error; LPIPS: learned perceptual image patch similarity; SSIM: structural similarity index.

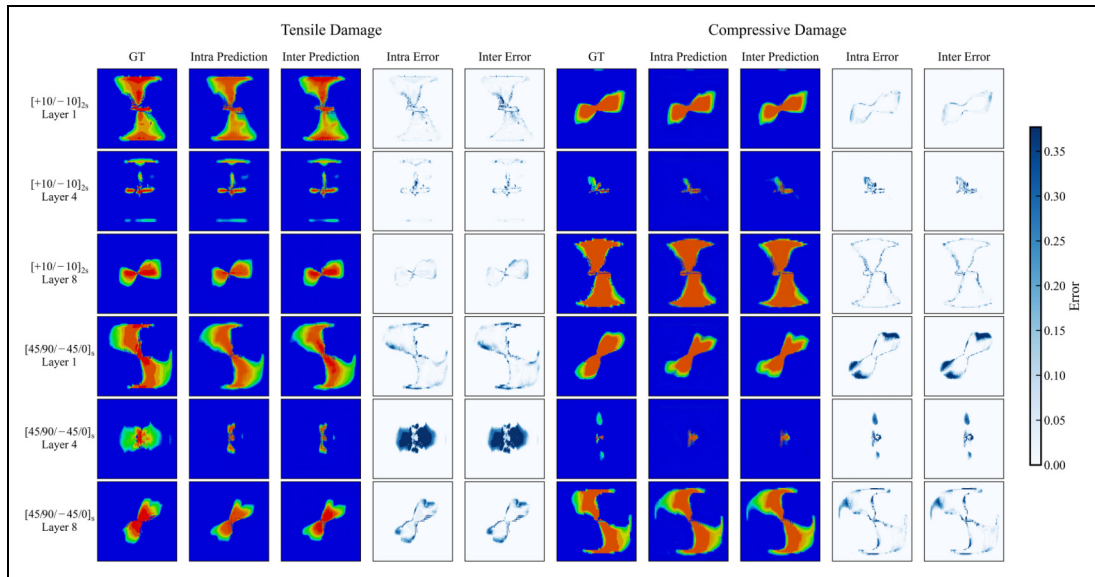


Figure A11. T700GC/M2I unseen-sequence prediction examples. ABAQUS ground truth, intramaterial prediction, intermaterial prediction, and their error maps are shown for T700GC/M2I unseen stacking sequences $[+10/-10]_{2s}$ and $[45/90/-45/0]_{4s}$. The selected layers check whether the two surrogate settings still follow the main damage zones when the lay-up is new. Both models reproduce the primary damaged regions. The intermaterial results are visually close to the intramaterial results and match the higher fiber compressive damage SSIM seen in the metric plot.

A.7 Mesh sensitivity analysis

A mesh sensitivity analysis is conducted to check the stability of the FE reference model. Four global mesh sizes, namely 1.5, 2.0, 2.5, and 3.0 mm, are compared using the contact-force history under the same representative impact condition. As shown in Figure A12, the 2.0 mm mesh gives a response close to the finer 1.5 mm mesh during the main loading stage, while the coarser 2.5 and 3.0 mm meshes show clear deviations in the peak value and the later force response. Therefore, the 2.0 mm mesh is adopted in the nonrefined region, together with the 0.8 mm refined central mesh, to balance response stability and computational cost for batch FE data generation. The adjacent plies were modeled using surface-to-surface contact. The local force drop observed with the 1.5 mm mesh is mainly attributed to mesh sensitivity in the timing and localization of damage softening and element deletion, together with the redistribution of the effective contact area and interlaminar load-transfer path during damage evolution. The 2.0 mm mesh preserves the main loading and unloading trend while providing a practical balance between response stability and computational cost.

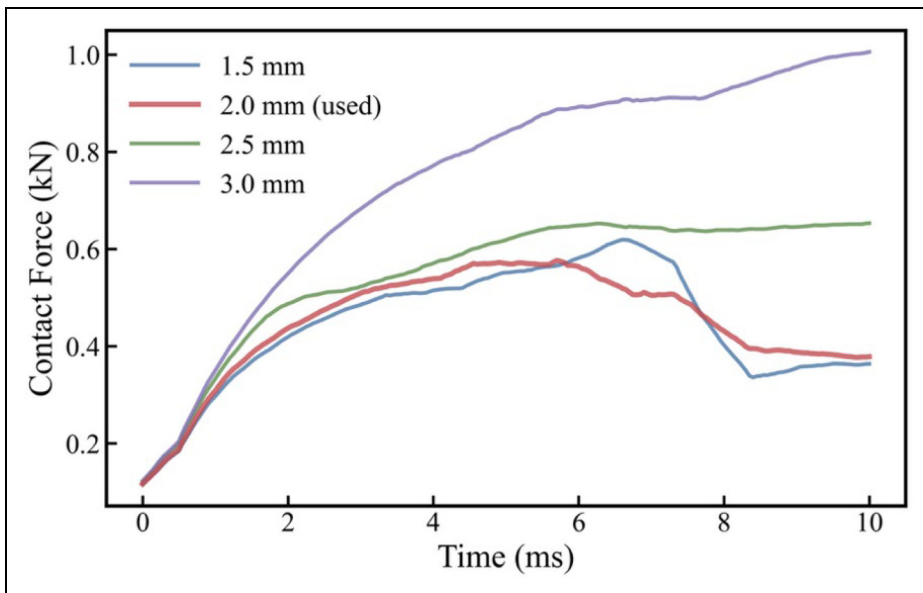


Figure A12. Mesh sensitivity analysis based on contact-force histories under a representative LVI condition. The 2.0 mm mesh used in the FE model gives a force response close to the finer 1.5 mm mesh during the main loading stage, while the coarser meshes show larger deviations. FE: finite element.

A.8 Codebook-size sensitivity analysis

To determine the codebook size, the DIR was retrained on the fiber tensile-damage dataset using $K = 256$, 512, and 1024. Increasing K improves RMSE, LPIPS, and SSIM, but the gain from 512 to 1024 is accompanied by a marked reduction in codebook utilization. This indicates that many additional embeddings remain underused, while increasing memory use and quantization cost. In contrast, $K = 256$ provides

insufficient representation capacity and gives poorer reconstruction. Therefore, $K = 512$ was selected as a balance between reconstruction accuracy and codebook efficiency. See Figure A13.

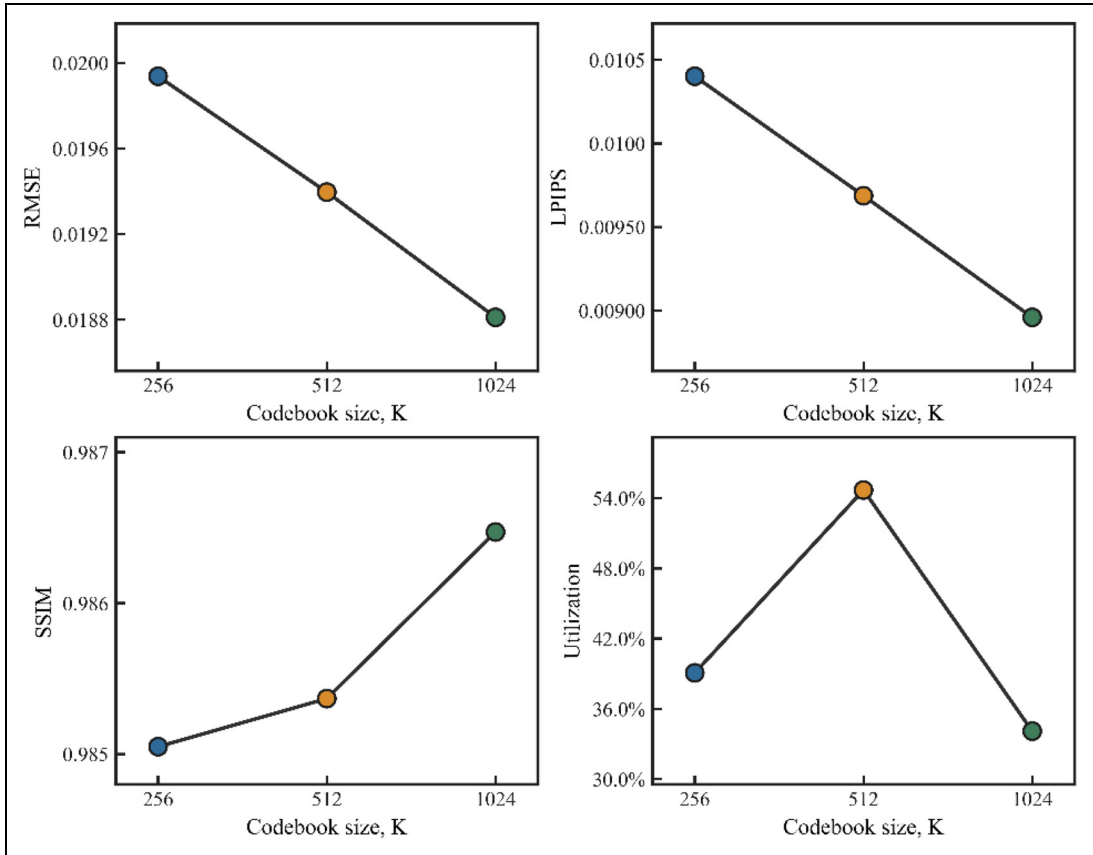


Figure A13. Codebook-size sensitivity analysis for the DIR trained on the fiber tensile-damage dataset. RMSE, LPIPS, SSIM, and codebook utilization are reported for $K = 256, 512$ and 1024 . RMSE: root mean square error; LPIPS: learned perceptual image patch similarity; SSIM: structural similarity index.

A.9 Force–time response and damage field extraction time

To clarify the time step used for damage field extraction, Figure A14 shows a representative impactor force–time response together with damage snapshots at selected increments. All damage fields used in the dataset were extracted at the final simulation time $t = 0.01$ s. By this stage, the impactor had entered the rebound phase in nearly all cases, and the damage pattern was close to its maximum extent. The same extraction criterion was applied to every simulation.

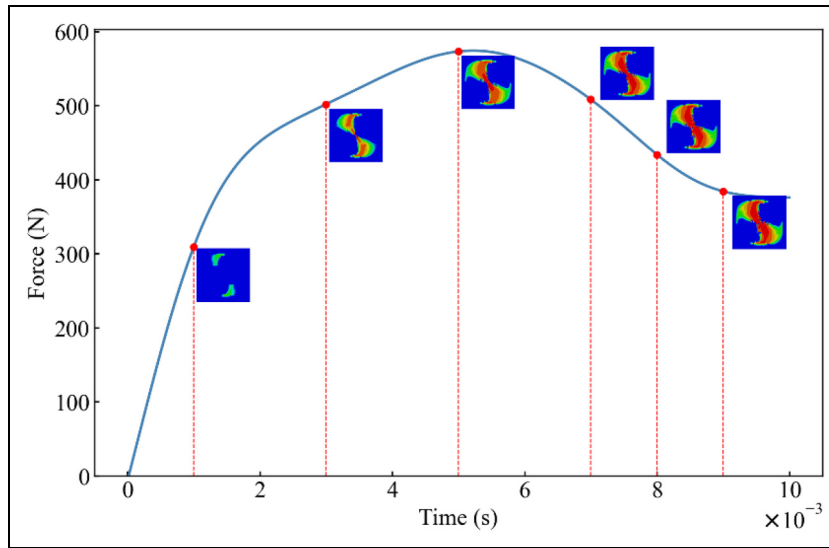


Figure A14. Representative force–time response and damage evolution during impact. Red markers indicate the increments at which the damage snapshots are shown. The final snapshot at $t = 0.01$ s corresponds to the extraction time used for the damage fields in Figure 5 and throughout the dataset.